

How Much Can Gaussian Dependence Inflate the Benjamini–Hochberg Procedure’s FDR?

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Abstract

We study the worst-case false discovery rate (FDR) of the Benjamini–Hochberg procedure for both one- and two-sided Gaussian tests when the correlation matrix is otherwise unrestricted. In each setting we construct finite Gaussian models for which the inflation factor $\text{FDR}(\text{BH}_q)/q$ diverges as $q \downarrow 0$, disproving any universal multiplicative FDR bound. For two-sided tests, the supremum over the number of hypotheses, mean vector, and correlation matrix is at least an explicit $\ell_=(q) > q$ satisfying

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q), \quad c_\ell = 0.6492828\dots$$

For the one-sided hypotheses $H_i : \theta_i \leq 0$, a sign-reversed common-factor construction gives the stronger explicit lower bound

$$\ell_\leq(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + \bar{\Phi}\left(\sqrt{2\log(1/q)}\right) > q,$$

with $\ell_\leq(q) = q\sqrt{\log(1/q)}/\sqrt{\pi} + q/2 + o(q)$. The nulls are positively correlated, whereas null–nonnull correlations are negative. Finally, we prove an $O\{q\sqrt{\log(1/q)}\}$ upper bound for the two-sided common-factor class and the matching one-sided upper bound $q\sqrt{\log(1/q)}/\sqrt{\pi} + O(q)$.

1 Introduction

We observe a Gaussian vector $T = (T_1, \dots, T_m)$ with mean θ and covariance matrix Σ , where $\Sigma_{ii} = 1$ for all i . We consider two testing problems. For two-sided tests, $H_i : \theta_i = 0$ and

$$P_i = p(|T_i|), \quad p(x) = 2\bar{\Phi}(x) = 2\{1 - \Phi(x)\}.$$

For one-sided tests, $H_i : \theta_i \leq 0$ and

$$P_i = p_+(T_i), \quad p_+(x) = \bar{\Phi}(x) = 1 - \Phi(x).$$

Here Φ and $\phi = \Phi'$ are the standard normal CDF and density. We use the conventions $p^{-1}(0) = p_+^{-1}(0) = \infty$. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] at level q orders the p -values as

$$P_{(1)} \leq \dots \leq P_{(m)}$$

and rejects the hypotheses corresponding to

$$P_{(1)}, \dots, P_{(R)}, \quad R = \max\{1 \leq k \leq m : P_{(k)} \leq qk/m\},$$

with $R = 0$ if the set is empty. If V is the number of rejected true nulls, then the false discovery proportion (FDP) and false discovery rate (FDR) are

$$\text{FDP} = \frac{V}{R \vee 1}, \quad \text{FDR} = \mathbb{E}(\text{FDP}).$$

A long-standing conjecture in the FDR literature is that BH controls the FDR at its nominal level q , regardless of the correlation matrix [Reiner-Benaim, 2007, Benjamini, 2010, Roux, 2018, Sarkar, 2023, Sarkar and Zhang, 2025]. Benjamini [2010, p. 407] summarized the evidence as follows:

“The modification to general dependence is often not needed: convincing simutheoretical evidence indicates that the same holds for two-sided z -tests with any correlation structure.”

Benjamini then noted that a complete theoretical proof was still missing. These works stated or supported this conjecture. However, recently, Dobriban [2026] found a counterexample where $\text{FDR} > 0.0104$ when $q = 0.01$.

In this paper, we ask a more refined question in each testing problem: how large can the FDR be as a function of q ? For two-sided tests, we seek a lower bound $\ell_=(q)$ such that

$$\sup_{\substack{m \in \mathbb{Z}_+, \theta \in \mathbb{R}^m, \Sigma \in \mathbb{S}_+^m \\ \Sigma_{ii}=1, 1 \leq i \leq m}} \text{FDR}(\text{BH}_q) \geq \ell_=(q). \quad (1)$$

Here $\mathbb{Z}_+ = \{1, 2, \dots\}$ and $\mathbb{S}_+^m$ denotes the cone of $m \times m$ symmetric positive semidefinite matrices. A relaxed folklore conjecture is that the worst-case FDR in (1) is bounded above by Cq for some universal constant $C > 0$. However, Proposition 2.2 shows that the two-sided lower bound constructed here satisfies, as $q \downarrow 0$,

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q), \quad c_\ell = 0.6492828\dots$$

In particular, the FDR inflation factor FDR/q can diverge as $q \downarrow 0$.

Moreover, we derive an exact formula for $\ell_=(q)$. We prove that $\ell_=(q) > q$ for every $q \in (0, 1)$. Instead of seeking a counterexample for a given nominal level q , our result disproves the conjecture that BH controls the FDR for correlated two-sided Gaussian tests. We plot the absolute deviation $\ell_=(q) - q$ and relative inflation $\ell_=(q)/q$ in Figure 1.

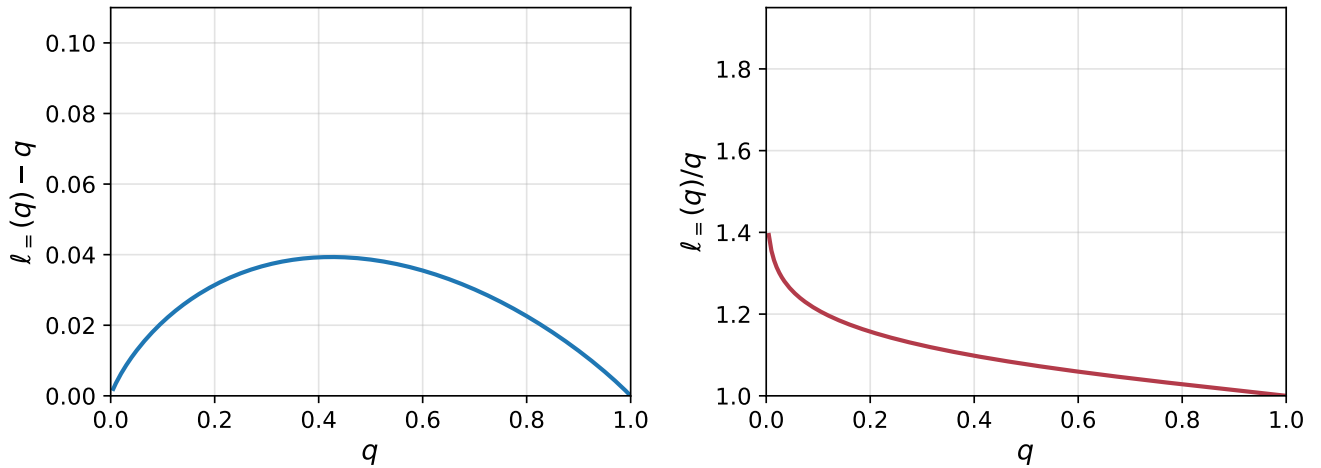


Figure 1: Numerical evaluation of $\ell_=(q) - q$ and $\ell_=(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 200$.

For one-sided Gaussian tests, the classical negatively correlated examples of Hochberg and Rom [1995] and Samuel-Cahn [1996] yield only small inflation. Chi et al. [2025] prove useful upper bounds

under several notions of negative dependence, but the adversarial construction used in their sharpness comparison, due to Su [2018], is not jointly Gaussian. Our one-sided construction instead retains positively correlated nulls and reverses the loading of the nonnulls. It gives, for every $q \in (0, 1)$,

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + \bar{\Phi}\left(\sqrt{2\log(1/q)}\right) > q,$$

and

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

This is a larger leading constant than in the two-sided construction and again makes FDR/q diverge.

Figure 2 plots $\ell_{\leq}(q) - q$ and $\ell_{\leq}(q)/q$ for $q = 0.005k$, $k = 1, \dots, 200$.

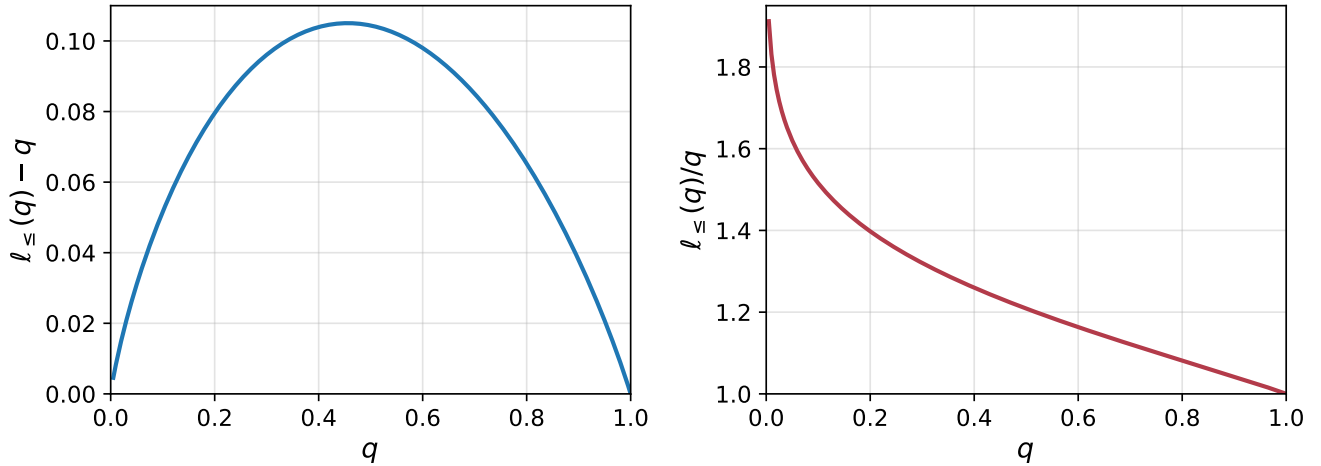


Figure 2: Numerical evaluation of $\ell_{\leq}(q) - q$ and $\ell_{\leq}(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 200$.

The dependence mechanisms differ sharply between the two constructions. Every covariance in the two-sided construction is positive, but the map $T_i \mapsto 2\bar{\Phi}(|T_i|)$ is not monotone and need not preserve positive regression dependence on a subset (PRDS). For one-sided Gaussian p -values, by contrast, nonnegative correlations between every true null and every coordinate imply PRDS and hence FDR control [Benjamini and Yekutieli, 2001]. Negative dependence touching a true null is therefore essential in the one-sided construction: null–null correlations are positive, while null–nonnull correlations are negative. Thus, the vector is not PRDS.

The lower-bound rate $q\sqrt{\log(1/q)}$ should be compared with the FDR-linking theorem of Su [2018]. Positive regression dependence within nulls (PRDN) relaxes PRDS by imposing the relevant positive dependence condition only on the null p -values. Under PRDN, the FDR-linking argument gives an $O(q\log(1/q))$ bound. Section 4 proves sharper Gaussian bounds. For two-sided one-common-factor models,

$$\text{FDR}(\text{BH}_q) = O(q\sqrt{\log(1/q)}),$$

uniformly over the number of hypotheses, means, and loadings. Thus, in this common-factor Gaussian class, the upper-bound rate improves on the general PRDN linking bound and matches the asymptotic order of $\ell_{\leq}(q)$ as $q \downarrow 0$. Moreover, as explained in Remark 4.2, the two-sided null p -values in this class satisfy the PRDN property by the absolute-value Gaussian MTP_2 criterion of Karlin and Rinott [1981]. Hence the upper bound sharpens the general PRDN-based linking rate within this common-factor subclass. For one-sided rank-one models, Theorem 4.3 gives an upper bound with leading term $q\sqrt{\log(1/q)}/\sqrt{\pi}$, and Corollary 4.4 shows that this matches the leading constant in $\ell_{\leq}(q)$. Both bounds are uniform over the number of hypotheses, means, and scalar loadings.

2 Lower bound for two-sided tests

2.1 Main result

$$M(y) = \sup_{a \geq 0} e^{-a^2/2} \cosh(ay). \quad (2)$$

Appendix A identifies the likelihood-ratio envelope: $M = 1$ on $[0, 1]$, and M is continuous, strictly increasing, and unbounded on $(1, \infty)$. Hence there is a unique $y_q > 1$ such that

$$qM(y_q) = 1. \quad (3)$$

Set

$$\ell_=(q) = q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q). \quad (4)$$

The next theorem shows that the lower bound can be approached arbitrarily closely by finite-dimensional Gaussian models with positive definite correlation matrices.

Theorem 2.1. *For $T \sim N(\theta, \Sigma)$, let $\text{FDR}_{\theta, \Sigma}(\text{BH}_q)$ denote the FDR of BH_q applied to T . For every $q \in (0, 1)$ and $\delta > 0$, there are $N < \infty$, $\theta \in \mathbb{R}^N$, and a positive definite correlation matrix Σ such that*

$$\text{FDR}_{\theta, \Sigma}(\text{BH}_q) > \ell_=(q) - \delta. \quad (5)$$

Consequently,

$$\sup_{\substack{N \in \mathbb{N}, \theta \in \mathbb{R}^N, \\ \Sigma \succ 0, \text{diag}(\Sigma) = \mathbf{1}_N}} \text{FDR}_{\theta, \Sigma}(\text{BH}_q) \geq \ell_=(q). \quad (6)$$

The next proposition gives the small- q expansion of the lower bound.

Proposition 2.2. *As $q \downarrow 0$,*

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q),$$

where

$$\begin{aligned} c_\ell &= \Phi(1) - \frac{1}{2\sqrt{2\pi}} + \int_1^\infty \left\{ M(y)\phi(y) - \frac{1}{2\sqrt{2\pi}} \right\} dy \\ &= 0.6492828 \dots \end{aligned}$$

Proposition 2.3 further shows that $\ell_=(q) > q$ for every $q \in (0, 1)$. Consequently, BH fails to control the FDR at every nontrivial nominal level in the correlated two-sided Gaussian setting.

Proposition 2.3. *For every $q \in (0, 1)$,*

$$\ell_=(q) > q.$$

Figure 1 in the Introduction plots $\ell_=(q) - q$ and $\ell_=(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 200$. Table 1 gives the absolute inflation $\ell_=(q) - q$ and relative inflation $\ell_=(q)/q$ for commonly used nominal levels.

q	0.01	0.05	0.10	0.20
$\ell_=(q)$	0.0136	0.0629	0.1209	0.2314
$\ell_=(q)/q$	1.3553	1.2571	1.2094	1.1570

Table 1: The lower bound $\ell_=(q)$ and relative inflation $\ell_=(q)/q$ at commonly used nominal levels.

2.2 Roadmap

Fix $q \in (0, 1)$ and a design law ν_q . Let $\Omega_N = ((\Theta_1, B_1), \dots, (\Theta_N, B_N))$, where the pairs are i.i.d. draws from ν_q , independently of $Z, \varepsilon_1, \dots, \varepsilon_N \stackrel{\text{i.i.d.}}{\sim} N(0, 1)$, and set

$$T_i = \Theta_i + B_i Z + \sqrt{1 - B_i^2} \varepsilon_i, \quad i = 1, \dots, N.$$

Conditional on a realization of Ω_N , this is a finite deterministic Gaussian design. Since BH is a step-up procedure, its rejection set is $\{i : |T_i| \geq \hat{C}_N\}$. We will show in Subsection 2.4 that, conditional on $Z = z$, $\hat{C}_N \rightarrow C(z)$ and $\text{FDP}_N \rightarrow D(z)$. Consequently,

$$\mathbb{E}_{\Omega_N}[\text{FDR}_{\Omega_N}(\text{BH}_q)] = \mathbb{E}_{\Omega_N, Z, \varepsilon}[\text{FDP}_N] \longrightarrow \mathbb{E}[D(Z)].$$

Thus, for any $\delta > 0$ and all sufficiently large N , at least one deterministic realization ω_N of Ω_N satisfies

$$\text{FDR}_{\omega_N}(\text{BH}_q) \geq \mathbb{E}[D(Z)] - \delta.$$

We next construct a family of laws $\nu_{q,r}$. Write D_r for the conditional FDP limit associated with $\nu_{q,r}$. For every $z \notin \{-1, -y_q\}$, we prove that $D_r(z) \rightarrow D_*(z)$ as $r \downarrow 0$, where

$$D_*(z) = \begin{cases} q, & z \geq -1, \\ qM(-z), & -y_q < z < -1, \\ 1, & z \leq -y_q. \end{cases}$$

Because $0 \leq D_r, D_* \leq 1$, dominated convergence then gives

$$\mathbb{E}[D_r(Z)] \longrightarrow \mathbb{E}[D_*(Z)] = \ell_=(q).$$

The argument therefore takes the iterated limit $\lim_{r \downarrow 0} \lim_{N \rightarrow \infty}$: first $N \rightarrow \infty$ for each fixed r , and then $r \downarrow 0$. To obtain a single finite example, we choose r sufficiently small and then N sufficiently large. A small independent Gaussian perturbation finally makes its covariance matrix positive definite.

2.3 Population quantities

Put $u_q = p^{-1}(q)$. Let ν_q be a probability measure on $\mathbb{R} \times [-1, 1]$. Write

$$\pi_0 = \nu_q(\{0\} \times [-1, 1]), \tag{7}$$

and assume $\pi_0 > 0$.

For $(\theta, b) \in \mathbb{R} \times [-1, 1]$, $z \in \mathbb{R}$, and $s \in (0, 1]$, let $u_s = p^{-1}(s)$ and define

$$F_{\theta, b, z}(s) = \mathbb{P}\{p(|T_i|) \leq s \mid \Theta_i = \theta, B_i = b, Z = z\}.$$

If $|b| < 1$, then

$$F_{\theta, b, z}(s) = \bar{\Phi}\left(\frac{u_s - \theta - bz}{\sqrt{1 - b^2}}\right) + \bar{\Phi}\left(\frac{u_s + \theta + bz}{\sqrt{1 - b^2}}\right). \tag{8}$$

If $|b| = 1$, then

$$F_{\theta, b, z}(s) = \mathbf{1}\{p(|\theta + bz|) \leq s\}. \tag{9}$$

Set $F_{\theta, b, z}(0) = 0$.

Conditional on $Z = z$, the CDFs of the mixture p -value and the null p -value are

$$R_z(s) = \int F_{\theta, b, z}(s) \nu_q(d\theta, db), \tag{10}$$

$$R_{0, z}(s) = \frac{1}{\pi_0} \int_{\{0\} \times [-1, 1]} F_{0, b, z}(s) \nu_q(d\theta, db). \tag{11}$$

For $c \geq u_q$, define the normalized quantities

$$\bar{R}_z(c) = \frac{qR_z\{p(c)\}}{p(c)}, \quad (12)$$

$$\bar{R}_{0,z}(c) = \frac{q\pi_0 R_{0,z}\{p(c)\}}{p(c)}. \quad (13)$$

Then the limiting cutoff can be written as

$$C(z) = \inf\{c \geq u_q : \bar{R}_z(c) \geq 1\}. \quad (14)$$

Here and below, $\inf \emptyset = \infty$. The limiting conditional FDP is defined by

$$D(z) = \begin{cases} \bar{R}_{0,z}\{C(z)\}, & C(z) < \infty, \\ 0, & C(z) = \infty. \end{cases} \quad (15)$$

We call a finite cutoff a regular first crossing when it satisfies the following conditions.

Definition 2.4. A finite cutoff $C(z) > u_q$ is regular if

$$\sup_{u_q \leq c \leq C(z) - \rho} \bar{R}_z(c) < 1 \quad \text{for every } 0 < \rho < C(z) - u_q, \quad (16)$$

$$\sup_{C(z) < c < C(z) + \rho} \bar{R}_z(c) > 1 \quad \text{for every } \rho > 0, \quad (17)$$

and $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$.

The following proposition reduces regularity to a local monotonicity check.

Proposition 2.5. Let $C(z) > u_q$ be finite. Suppose there exist $u_q < a < C(z) < b$ such that the map $c \mapsto \bar{R}_z(c)$ is differentiable on (a, b) and satisfies

$$\bar{R}'_z(c) > 0, \quad a < c < b.$$

Suppose also that $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$. Then $C(z)$ is regular.

Proof. The map $c \mapsto R_z\{p(c)\}$ is a survival function and hence is upper semicontinuous. Since p is positive and continuous, $c \mapsto \bar{R}_z(c)$ is also upper semicontinuous. By the definition of $C(z)$, $\bar{R}_z(c) < 1$ for every $c < C(z)$. Therefore, for every $0 < \rho < C(z) - u_q$, upper semicontinuity and compactness give

$$\sup_{u_q \leq c \leq C(z) - \rho} \bar{R}_z(c) < 1,$$

which proves (16).

There is a sequence $c_n \geq C(z)$ decreasing to $C(z)$ such that $\bar{R}_z(c_n) \geq 1$. Continuity on (a, b) therefore gives $\bar{R}_z\{C(z)\} = 1$. The mean value theorem and the derivative condition imply strict monotonicity on (a, b) , which yields

$$\bar{R}_z(c) > 1 \quad \text{for } C(z) < c < b.$$

This gives (17). Continuity of the null curve at $C(z)$ completes Definition 2.4. $\square$

2.4 Limiting cutoff and FDR

Let

$$\Omega_N = ((\Theta_1, B_1), \dots, (\Theta_N, B_N)), \quad (\Theta_i, B_i) \stackrel{\text{i.i.d.}}{\sim} \nu_q. \quad (18)$$

Since the distribution of $(T_1, \dots, T_N)$ is fully determined by Ω_N , we write $\text{FDR}_{\theta, \Sigma}(\text{BH}_q)$ as $\text{FDR}_{\Omega_N}(\text{BH}_q)$.

Define $P_i = p(|T_i|)$.

$$\hat{R}_N(s) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{P_i \leq s\}, \quad (19)$$

$$\hat{R}_{0,N}(s) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\Theta_i = 0, P_i \leq s\}. \quad (20)$$

The following lemma gives a uniform conditional empirical-CDF bound.

Lemma 2.6. *For every $z \in \mathbb{R}$ and $t > 0$,*

$$\mathbb{P}\left(\max\left\{\|\hat{R}_N - R_z\|_\infty, \|\hat{R}_{0,N} - \pi_0 R_{0,z}\|_\infty\right\} > t \mid Z = z\right) \leq 4e^{-2Nt^2}. \quad (21)$$

Proof. Conditional on $Z = z$, the P_i are i.i.d. with CDF R_z . Also define

$$W_i = P_i \mathbf{1}\{\Theta_i = 0\} + 2\mathbf{1}\{\Theta_i \neq 0\}.$$

For $0 \leq s \leq 1$,

$$\mathbf{1}\{W_i \leq s\} = \mathbf{1}\{\Theta_i = 0, P_i \leq s\}, \quad \mathbb{P}(W_i \leq s \mid Z = z) = \pi_0 R_{0,z}(s).$$

Apply the Dvoretzky–Kiefer–Wolfowitz inequality twice and take a union bound. $\square$

The next lemma gives an exact characterization of the BH cutoff and FDP through the empirical CDFs.

Lemma 2.7. *Let R_N be the number of BH rejections and put*

$$\hat{\tau}_N = \frac{qR_N}{N}. \quad (22)$$

If $R_N \geq 1$, then

$$\hat{R}_N(\hat{\tau}_N) = \frac{R_N}{N}, \quad \text{FDP}_N = \frac{q\hat{R}_{0,N}(\hat{\tau}_N)}{\hat{\tau}_N}. \quad (23)$$

With $\hat{C}_N = p^{-1}(\hat{\tau}_N)$,

$$\hat{C}_N = \inf \left\{ c \geq u_q : \frac{q\hat{R}_N\{p(c)\}}{p(c)} \geq 1 \right\}. \quad (24)$$

Then the BH rejection set is $\{i : |T_i| \geq \hat{C}_N\}$. If $R_N = 0$, the set in (24) is empty and $\hat{C}_N = \infty$.

Proof. Let $P_{(1)} \leq \dots \leq P_{(N)}$. If more than R_N values were at most qR_N/N , then

$$P_{(R_N+1)} \leq qR_N/N < q(R_N+1)/N,$$

contradicting the maximality of R_N . Hence $N\hat{R}_N(\hat{\tau}_N) = R_N$, and (23) follows. Since p is decreasing, the BH step-up definition is equivalent to (24). $\square$

The next theorem establishes the limiting cutoff conditional on the common factor Z .

Theorem 2.8. Assume that $C(z)$ is regular for almost every z . Then, for every z at which $C(z)$ is regular, conditional on $Z = z$,

$$\hat{C}_N \longrightarrow C(z), \quad \text{FDP}_N \longrightarrow D(z) = \bar{R}_{0,z}\{C(z)\} = \frac{q\pi_0 R_{0,z}[p\{C(z)\}]}{p\{C(z)\}} \quad (25)$$

in probability. Consequently,

$$\lim_{N \rightarrow \infty} \mathbb{E}_{\Omega_N} [\text{FDR}_{\Omega_N}(\text{BH}_q)] = \mathbb{E}[D(Z)], \quad (26)$$

where

$$\text{FDR}_{\Omega_N}(\text{BH}_q) = \mathbb{E}_{Z,\varepsilon} [\text{FDP}_N \mid \Omega_N]. \quad (27)$$

Proof. Fix a regular z and $\rho, \tau > 0$. Since $D(z) = \bar{R}_{0,z}\{C(z)\}$ and $c \mapsto \bar{R}_{0,z}(c)$ is continuous at $C(z)$, choose

$$0 < \rho(z) < \min\{\rho, C(z) - u_q\}$$

so that

$$\sup_{C(z)-\rho(z) \leq c \leq C(z)+\rho(z)} |\bar{R}_{0,z}(c) - D(z)| \leq \tau. \quad (28)$$

Set $A(z) = C(z) - \rho(z)$. The left condition in Definition 2.4 gives $\sup_{u_q \leq c \leq A(z)} \bar{R}_z(c) < 1$, and its right condition permits a choice of $B(z) \in (C(z), C(z) + \rho(z))$ with $\bar{R}_z\{B(z)\} > 1$. Set

$$\gamma(z) = \frac{1}{2} \min \left\{ 1 - \sup_{u_q \leq c \leq A(z)} \bar{R}_z(c), \quad \bar{R}_z\{B(z)\} - 1 \right\} > 0.$$

Put

$$E_N(z) = \max \left\{ \|\hat{R}_N - R_z\|_\infty, \|\hat{R}_{0,N} - \pi_0 R_{0,z}\|_\infty \right\}.$$

Lemma 2.6 gives $E_N(z) \rightarrow 0$ in probability conditional on $Z = z$. For every $u_q \leq c \leq B(z)$,

$$\begin{aligned} \left| \frac{q\hat{R}_N\{p(c)\}}{p(c)} - \bar{R}_z(c) \right| &\leq \frac{qE_N(z)}{p\{B(z)\}}, \\ \left| \frac{q\hat{R}_{0,N}\{p(c)\}}{p(c)} - \bar{R}_{0,z}(c) \right| &\leq \frac{qE_N(z)}{p\{B(z)\}}. \end{aligned} \quad (29)$$

By the empirical curve we mean

$$c \mapsto \frac{q\hat{R}_N\{p(c)\}}{p(c)}.$$

On the event $qE_N(z)/p\{B(z)\} < \gamma(z)$, the definition of $\gamma(z)$ and (29) give

$$\begin{aligned} \sup_{u_q \leq c \leq A(z)} \frac{q\hat{R}_N\{p(c)\}}{p(c)} &\leq \sup_{u_q \leq c \leq A(z)} \bar{R}_z(c) + \frac{qE_N(z)}{p\{B(z)\}} \\ &< 1 - 2\gamma(z) + \gamma(z) < 1, \end{aligned}$$

whereas

$$\frac{q\hat{R}_N\{p(B(z))\}}{p(B(z))} \geq \bar{R}_z\{B(z)\} - \frac{qE_N(z)}{p\{B(z)\}} > 1 + 2\gamma(z) - \gamma(z) > 1.$$

Lemma 2.7 and (28) therefore give

$$A(z) < \hat{C}_N \leq B(z), \quad |\text{FDP}_N - D(z)| \leq \tau + \frac{qE_N(z)}{p\{B(z)\}}.$$

Because $\rho, \tau > 0$ were arbitrary, this proves (25) at every regular z .

For completeness, the quantities in the expectation can be taken to be measurable without making a measurable selection of $B(z)$. The maps $(z, c) \mapsto \bar{R}_z(c)$ and $(z, c) \mapsto \bar{R}_{0,z}(c)$ are jointly Borel measurable, and the first is left-continuous in c . Define

$$C^\circ(z) = \inf\{c \in \mathbb{Q} : c \geq u_q, \bar{R}_z(c) > 1\}.$$

At every regular z , left continuity and (17) imply $C^\circ(z) = C(z)$. Thus C has a measurable version. Moreover, for $c < C(z)$, $\bar{R}_{0,z}(c) \leq \bar{R}_z(c) < 1$; continuity of the null curve at the crossing gives $0 \leq D(z) \leq 1$. Hence

$$D^\circ(z) = \begin{cases} \min[1, \bar{R}_{0,z}\{C^\circ(z)\}], & C^\circ(z) < \infty, \\ 0, & C^\circ(z) = \infty, \end{cases}$$

is a measurable $[0, 1]$ -valued version of $D(z)$ at almost every z . Use these versions below. Conditional convergence in probability, together with boundedness, now yields

$$\mathbb{E}[|\text{FDP}_N - D(z)| \mid Z = z] \longrightarrow 0 \quad \text{for almost every } z.$$

Dominated convergence and the tower property give

$$\mathbb{E}_{\Omega_N, Z, \varepsilon}[|\text{FDP}_N - D(Z)|] \longrightarrow 0,$$

and a second application of the tower property yields (26). $\square$

Lastly, we show that Σ can be made positive definite by an independent Gaussian perturbation. For a fixed realization $\omega = ((\theta_i, b_i))_{i=1}^N$, let $\xi \sim N(0, I_N)$ be independent and define

$$T_i^{(\eta)} = \theta_i + \sqrt{1 - \eta^2} \left(b_i Z + \sqrt{1 - b_i^2} \varepsilon_i \right) + \eta \xi_i, \quad 0 < \eta < 1. \quad (30)$$

Then

$$\Sigma_{\omega, \eta} = (1 - \eta^2) [bb^\top + \text{diag}(1 - b_1^2, \dots, 1 - b_N^2)] + \eta^2 I_N. \quad (31)$$

The following lemma shows that the perturbation makes the correlation matrix positive definite while preserving the FDR in the limit.

Lemma 2.9. *For every fixed ω ,*

$$\text{diag}(\Sigma_{\omega, \eta}) = \mathbf{1}_N, \quad \Sigma_{\omega, \eta} \succeq \eta^2 I_N, \quad (32)$$

and

$$\text{FDR}_\omega^{(\eta)}(\text{BH}_q) \longrightarrow \text{FDR}_\omega(\text{BH}_q) \quad (\eta \downarrow 0). \quad (33)$$

Proof. Equation (32) is immediate from (31). Under the coupling (30), $T^{(\eta)} \rightarrow T$ almost surely. Define $P_i^{(\eta)} = p(|T_i^{(\eta)}|)$. Then $P_i^{(\eta)} \rightarrow P_i = p(|T_i|)$ almost surely. Each P_i has a continuous marginal law, so

$$\mathbb{P} \left\{ P_i \in \left\{ \frac{qk}{N} : 1 \leq k \leq N \right\} \text{ for some } i \right\} = 0.$$

Off this event, the truth values of all finitely many comparisons $P_i^{(\eta)} \leq qk/N$ agree with those of $P_i \leq qk/N$ for all sufficiently small η . Hence the BH rejection count and rejected-null set are eventually constant. Bounded convergence proves (33). $\square$

2.5 Construction of ν_q

We construct a family of laws $\nu_{q,r}$ indexed by $r \downarrow 0$. For $a, y \geq 0$, put

$$L_a(y) = e^{-a^2/2} \cosh(ay). \quad (34)$$

For $y > 1$, let

$$a(y) = \arg \max_{\alpha \geq 0} L_\alpha(y), \quad y(\alpha) = a^{-1}(\alpha), \quad \alpha > 0. \quad (35)$$

Set

$$a_q = a(y_q). \quad (36)$$

Appendix A establishes the properties of these functions that we use below.

The quantities $b_r, \tau_r, \mu_r, \pi_{0,r}$, and c_r are defined below. The law $\nu_{q,r}$ has the three components displayed in Table 2.

Table 2: Two-sided population construction defining $\nu_{q,r}$.			
Type	Mass	Θ	B
Nulls	$\pi_{0,r}$	0	r
Primary nonnulls	r	$c_r + y(b_r)$	1
Secondary nonnulls	$\tau_r \mu_r(\mathrm{d}a)$	$a/r + y(a)$	1, $a \in [b_r, a_q]$

Equivalently, for every bounded measurable $f : \mathbb{R} \times [-1, 1] \rightarrow \mathbb{R}$,

$$\int f \, \mathrm{d}\nu_{q,r} = \pi_{0,r} f(0, r) + r f(c_r + y(b_r), 1) + \tau_r \int_{b_r}^{a_q} f\left(\frac{a}{r} + y(a), 1\right) \mu_r(\mathrm{d}a). \quad (37)$$

Definition of secondary non-nulls. For small r , set

$$b_r = r p^{-1}(r^2). \quad (38)$$

For the remaining definitions, take r small enough that $0 < b_r < a_q$. For $a \in [b_r, a_q]$, define

$$W_r(a) = \frac{[1 - qM\{y(a)\}]p(a/r)}{q}. \quad (39)$$

By Lemma A.1 and (167), the factor $1 - qM\{y(a)\}$ is nonnegative and decreasing on $[b_r, a_q]$; the factor $p(a/r)$ is decreasing in a . Hence W_r is continuous and decreasing, with $W_r(a_q) = 0$. Set

$$\tau_r := W_r(b_r) = \frac{[1 - qM\{y(b_r)\}]r^2}{q} = O(r^2). \quad (40)$$

We therefore call this the secondary-non-null block; its total mass is $O(r^2) = o(r)$. Let μ_r be the probability measure on $[b_r, a_q]$ determined by

$$\mu_r([a, a_q]) = \frac{W_r(a)}{\tau_r}, \quad b_r \leq a \leq a_q. \quad (41)$$

In particular, $\mu_r([b_r, a_q]) = W_r(b_r)/\tau_r = 1$. Equivalently,

$$\mu_r((a, b]) = \frac{W_r(a) - W_r(b)}{\tau_r}, \quad b_r \leq a < b \leq a_q. \quad (42)$$

Definition of nulls. Put

$$\pi_{0,r} = 1 - r - \tau_r.$$

Since $\pi_{0,r} + r + \tau_r = 1$, the masses in (37) define a probability measure.

Definition of primary non-nulls. For all sufficiently small r , $\pi_{0,r} > 0$. Define c_r by

$$p(c_r) = \frac{q(r + \tau_r)}{1 - q\pi_{0,r}}. \quad (43)$$

The numerator is positive. Since $\tau_r = O(r^2)$ and $\pi_{0,r} = 1 - r - \tau_r$,

$$1 - q\pi_{0,r} = 1 - q + qr + q\tau_r \longrightarrow 1 - q > 0, \quad q(r + \tau_r) \longrightarrow 0.$$

Thus the right side of (43) is positive and smaller than one for small r , so c_r is well-defined.

2.6 Population FDR analysis under $\nu_{q,r}$

2.6.1 Preliminaries

Apply Subsection 2.3 with $\nu_q = \nu_{q,r}$. Denote the resulting quantities by

$$R_{r,z}, \quad R_{0,r,z}, \quad \bar{R}_{r,z}, \quad \bar{R}_{0,r,z}, \quad C_r(z).$$

For $c \geq u_q$,

$$\begin{aligned} \bar{R}_{r,z}(c) &= q\pi_{0,r} \frac{R_{0,r,z}\{p(c)\}}{p(c)} \\ &\quad + \frac{qr}{p(c)} \mathbf{1}\{|c_r + y(b_r) + z| \geq c\} \\ &\quad + \frac{q\tau_r}{p(c)} \int_{b_r}^{a_q} \mathbf{1}\left\{\left|\frac{a}{r} + y(a) + z\right| \geq c\right\} \mu_r(da), \end{aligned} \quad (44)$$

and

$$\bar{R}_{0,r,z}(c) = q\pi_{0,r} \frac{R_{0,r,z}\{p(c)\}}{p(c)}. \quad (45)$$

Whenever $C_r(z)$ is regular, set

$$D_r(z) = \bar{R}_{0,r,z}\{C_r(z)\}. \quad (46)$$

Set $D_r(z) = 0$ otherwise. At every regular crossing, $0 \leq D_r(z) \leq 1$.

We state two lemmas that will be applied in the following analysis. The first lemma gives the order of b_r and c_r as $r \downarrow 0$.

Lemma 2.10. *As $r \downarrow 0$,*

$$b_r \downarrow 0, \quad \frac{b_r}{r} = p^{-1}(r^2) \rightarrow \infty, \quad p(b_r/r) = r^2, \quad y(b_r) \downarrow 1. \quad (47)$$

Moreover,

$$p(c_r) \sim \frac{qr}{1-q}, \quad c_r \rightarrow \infty, \quad rc_r \rightarrow 0, \quad \frac{b_r}{r} - c_r \rightarrow \infty. \quad (48)$$

Proof. The identity $p(b_r/r) = r^2$ follows directly from $b_r = r p^{-1}(r^2)$. Applying (176) in Lemma A.2 gives

$$\frac{b_r}{r} = p^{-1}(r^2) \sim \sqrt{4 \log(1/r)} \rightarrow \infty, \quad b_r \sim 2r \sqrt{\log(1/r)} \rightarrow 0.$$

The last limit in (47) follows from Lemma A.1: $a(y)$ is continuous and strictly increasing from $(1, \infty)$ onto $(0, \infty)$, with $\lim_{y \downarrow 1} a(y) = 0$. Since $y(\cdot)$ is its inverse and $b_r \downarrow 0$, we have $y(b_r) \downarrow 1$.

By (43) and (40),

$$p(c_r) = \frac{q(r + \tau_r)}{1 - q\pi_{0,r}} \sim \frac{qr}{1 - q}.$$

Thus

$$\log \frac{1}{p(c_r)} = \log \frac{1}{r} + O(1).$$

Applying (176) in Lemma A.2 gives

$$c_r = p^{-1}\{p(c_r)\} \sim \sqrt{2 \log(1/r)}.$$

Hence $c_r \rightarrow \infty$ and $rc_r \rightarrow 0$. Combining this with

$$\frac{b_r}{r} = p^{-1}(r^2) \sim \sqrt{4 \log(1/r)}$$

gives

$$\frac{b_r}{r} - c_r \sim (2 - \sqrt{2}) \sqrt{\log(1/r)} \rightarrow \infty.$$

□

The next lemma gives a uniform approximation to the normalized conditional FDP contribution from nulls.

Lemma 2.11. *For every $A > 0$ and $T \geq 0$, as $r \downarrow 0$,*

$$\sup_{|z| \leq T} \sup_{0 \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right| \rightarrow 0. \quad (49)$$

Proof. For $0 < r < 1$, the null pair $(\Theta, B) = (0, r)$ gives

$$R_{0,r,z}\{p(c)\} = \mathbb{P}\{|rz + \sqrt{1 - r^2} \varepsilon| \geq c\}.$$

Write

$$x_{\pm} = \frac{c \mp rz}{\sqrt{1 - r^2}}.$$

Then

$$R_{0,r,z}\{p(c)\} = \bar{\Phi}(x_+) + \bar{\Phi}(x_-).$$

Fix $C > 0$. Uniformly for $|z| \leq T$ and $C \leq c \leq A/r$,

$$\begin{aligned} x_{\pm} - c &= c \left\{ \frac{1}{\sqrt{1 - r^2}} - 1 \right\} \mp \frac{rz}{\sqrt{1 - r^2}} \\ &= \frac{cr^2}{\sqrt{1 - r^2}\{1 + \sqrt{1 - r^2}\}} \mp \frac{rz}{\sqrt{1 - r^2}}. \end{aligned}$$

Therefore,

$$|x_{\pm} - c| \leq \frac{Ar}{\sqrt{1 - r^2}\{1 + \sqrt{1 - r^2}\}} + \frac{Tr}{\sqrt{1 - r^2}} = O_{A,T}(r).$$

Consequently, for all sufficiently small r , the shifts $x_{\pm} - c$ are uniformly bounded and $x_{\pm} \geq C/2$. The uniform expansion (175) in Lemma A.2 therefore gives remainders $\rho_{r,C,\pm}(c, z)$ such that

$$\begin{aligned} \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(c)} &= \frac{c}{x_{\pm}} \exp \left\{ -\frac{x_{\pm}^2 - c^2}{2} \right\} \{1 + \rho_{r,C,\pm}(c, z)\}, \\ -\frac{x_{\pm}^2 - c^2}{2} &= -\frac{(rc)^2}{2} \pm rcz + O_{A,T}(r^2). \end{aligned} \quad (50)$$

and

$$\lim_{C \rightarrow \infty} \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \max_{\pm} |\rho_{r,C,\pm}(c, z)| = 0. \quad (51)$$

For each fixed $C > 0$, $c/x_{\pm} \rightarrow 1$ uniformly for $|z| \leq T$ and $C \leq c \leq A/r$ as $r \downarrow 0$. To identify the leading term, put $t = rc$, so that $0 \leq t \leq A$. Then

$$\frac{1}{2} \left[\exp \left\{ -\frac{t^2}{2} + tz \right\} + \exp \left\{ -\frac{t^2}{2} - tz \right\} \right] = e^{-t^2/2} \cosh(tz) = L_t(|z|).$$

Because t and z range over compact sets, these exponential factors are uniformly bounded. Thus, after summing the two tails in (50) and using $p(c) = 2\bar{\Phi}(c)$, the errors from $c/x_{\pm} - 1$ and from the $O_{A,T}(r^2)$ term vanish uniformly for each fixed C . Consequently, there is a constant $K_{A,T} < \infty$, independent of C , such that, for

$$\eta(C) = \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right|,$$

we have

$$\eta(C) \leq K_{A,T} \limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{C \leq c \leq A/r} \max_{\pm} |\rho_{r,C,\pm}(c, z)|.$$

The right side tends to zero as $C \rightarrow \infty$ by (51). Hence $\eta(C) \rightarrow 0$.

It remains to control $0 \leq c \leq C$. On this range, $p(c) \geq p(C) > 0$. Also, uniformly over $0 \leq c \leq C$ and $|z| \leq T$,

$$R_{0,r,z}\{p(c)\} = \bar{\Phi}\left(\frac{c - rz}{\sqrt{1 - r^2}}\right) + \bar{\Phi}\left(\frac{c + rz}{\sqrt{1 - r^2}}\right) = p(c) + O_{C,T}(r),$$

because the two arguments differ from c by $O_{C,T}(r)$ and $\bar{\Phi}$ has bounded derivative. Hence

$$\sup_{|z| \leq T} \sup_{0 \leq c \leq C} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - 1 \right| \rightarrow 0.$$

Moreover, $L_{rc}(|z|) = e^{-(rc)^2/2} \cosh(rc|z|) \rightarrow 1$ uniformly for $|z| \leq T$ and $0 \leq c \leq C$. Thus

$$\limsup_{r \downarrow 0} \sup_{|z| \leq T} \sup_{0 \leq c \leq A/r} \left| \frac{R_{0,r,z}\{p(c)\}}{p(c)} - L_{rc}(|z|) \right| \leq \eta(C).$$

Letting $C \rightarrow \infty$ proves the claim. $\square$

2.6.2 The regime $z > -1$

The next lemma locates the crossing created by the primary non-null.

Lemma 2.12. *For every compact $K \subset (-1, \infty)$ and every $H > 0$, uniformly for $z \in K$ and $|h| \leq H$,*

$$\bar{R}_{r,z}\left(c_r + \frac{h}{c_r}\right) \rightarrow q + (1 - q)e^h. \quad (52)$$

Moreover, for every $h_0 > 0$,

$$\limsup_{r \downarrow 0} \sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} \bar{R}_{r,z}(c) \leq q + (1 - q)e^{-h_0} < 1. \quad (53)$$

Consequently,

$$C_r(z) = c_r + o(c_r^{-1}) \quad (54)$$

and

$$D_r(z) \rightarrow q, \quad (55)$$

uniformly on K . Moreover, for sufficiently small r , $C_r(z)$ is regular for every $z \in K$.

Proof. Proof of (52). Fix $H > 0$, take $z \in K$, and take $|h| \leq H$. The primary non-null has $B = 1$ and $\Theta = c_r + y(b_r)$, so conditional on $Z = z$ its value is $c_r + y(b_r) + z$. Since $K \subset (-1, \infty)$ is compact,

$$\eta_K := \inf_{z \in K} (1 + z) > 0.$$

By Lemma 2.10, $c_r \rightarrow \infty$ and $y(b_r) \downarrow 1$. For the fixed H , this implies $H/c_r \rightarrow 0$. Combining these limits, for sufficiently small r ,

$$|y(b_r) - 1| \leq \frac{\eta_K}{4}, \quad \frac{H}{c_r} \leq \frac{\eta_K}{4}. \quad (56)$$

For $z \in K$ and $|h| \leq H$, the definition of η_K gives $z \geq -1 + \eta_K$, while (56) gives $y(b_r) \geq 1 - \eta_K/4$ and $-h/c_r \geq -H/c_r \geq -\eta_K/4$. Therefore

$$y(b_r) + z - \frac{h}{c_r} \geq \left(1 - \frac{\eta_K}{4}\right) + (-1 + \eta_K) - \frac{\eta_K}{4} = \frac{\eta_K}{2}, \quad z \in K, \quad |h| \leq H.$$

Equivalently, $y(b_r) + z \geq h/c_r + \eta_K/2$. Hence the conditional value of the primary non-null satisfies

$$\begin{aligned} |c_r + y(b_r) + z| &= c_r + y(b_r) + z \\ &= c_r + \frac{h}{c_r} + \left\{ y(b_r) + z - \frac{h}{c_r} \right\} \\ &\geq c_r + \frac{h}{c_r} + \frac{\eta_K}{2} > c_r + \frac{h}{c_r}, \end{aligned} \quad (57)$$

so the primary non-null is rejected at cutoff $c_r + h/c_r$, uniformly for $z \in K$ and $|h| \leq H$.

Next consider a secondary non-null indexed by $a \in [b_r, a_q]$. Its conditional value at $Z = z$ is $a/r + y(a) + z$. Since $a \geq b_r$, we have $a/r \geq b_r/r$. Also, Lemma A.1 shows that $y(\cdot)$, the inverse of the strictly increasing map $a(\cdot)$, is increasing; hence $y(a) \geq y(b_r)$. Together with $z \geq \inf_{u \in K} u$ and $h \leq H$, this gives

$$\frac{a}{r} + y(a) + z - \left(c_r + \frac{h}{c_r}\right) \geq \left(\frac{b_r}{r} - c_r\right) + y(b_r) + \inf_{u \in K} u - \frac{H}{c_r}. \quad (58)$$

Here $\frac{b_r}{r} - c_r \rightarrow \infty$ by (48), while

$$y(b_r) + \inf_{u \in K} u - \frac{H}{c_r} \longrightarrow 1 + \inf_{u \in K} u \quad (59)$$

because $y(b_r) \downarrow 1$ and $H/c_r \rightarrow 0$. Equations (48) and (59) show that the right side of (58) tends to $+\infty$. Hence it is positive for sufficiently small r , uniformly in $z \in K$, $|h| \leq H$, and $a \in [b_r, a_q]$. Thus every secondary non-null is rejected at cutoff $c_r + h/c_r$.

Lastly, we analyze the nulls. Since $|h| \leq H$ and $rc_r \rightarrow 0$ by (48),

$$r \left(c_r + \frac{h}{c_r} \right) = rc_r + \frac{rh}{c_r} \longrightarrow 0. \quad (60)$$

Let $T_K = \sup_{z \in K} |z| < \infty$. Since $c_r \rightarrow \infty$ by (48) in Lemma 2.10, while (60) gives a vanishing rescaled cutoff, we have

$$u_q \leq c_r + \frac{h}{c_r} \leq \frac{1}{r}$$

for all sufficiently small r , uniformly for $z \in K$ and $|h| \leq H$. Applying (49) in Lemma 2.11 with $A = 1$ and $T = T_K$ gives

$$\frac{R_{0,r,z}\{p(c_r + h/c_r)\}}{p(c_r + h/c_r)} = L_{r(c_r + h/c_r)}(|z|) + o(1).$$

Moreover,

$$L_{r(c_r+h/c_r)}(|z|) = \exp \left\{ -\frac{r^2(c_r + h/c_r)^2}{2} \right\} \cosh(r(c_r + h/c_r)|z|) = 1 + o(1),$$

uniformly for $z \in K$ and $|h| \leq H$. Since $\pi_{0,r} = 1 - r - \tau_r \rightarrow 1$, it follows that

$$q\pi_{0,r} \frac{R_{0,r,z}\{p(c_r + h/c_r)\}}{p(c_r + h/c_r)} = q + o(1).$$

By (43),

$$\frac{q(r + \tau_r)}{p(c_r + h/c_r)} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p(c_r + h/c_r)} \rightarrow (1 - q)e^h$$

by (180) in Lemma A.2. This proves (52).

Proof of (53). Fix $h_0 > 0$. Equation (57), evaluated at $h = h_0$, gives

$$|c_r + y(b_r) + z| > c_r + \frac{h_0}{c_r},$$

so the primary non-null is rejected at cutoff $c_r + h_0/c_r$. Equations (58), (48), and (59) give

$$\frac{a}{r} + y(a) + z > c_r + \frac{h_0}{c_r}, \quad a \in [b_r, a_q],$$

for sufficiently small r , uniformly in $z \in K$. Thus every secondary non-null is also rejected at cutoff $c_r + h_0/c_r$. Since rejection at cutoff c means $|T_i| \geq c$, all non-nulls are rejected for every $u_q \leq c \leq c_r + h_0/c_r$. Therefore, on this interval,

$$\bar{R}_{r,z}(c) = q\pi_{0,r} \frac{R_{0,r,z}\{p(c)\}}{p(c)} + \frac{q(r + \tau_r)}{p(c)}.$$

For $u_q \leq c \leq c_r - h_0/c_r$, Lemma 2.11 and $rc_r \rightarrow 0$ from (48) give

$$\sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} q\pi_{0,r} \frac{R_{0,r,z}\{p(c)\}}{p(c)} \leq q + o(1).$$

Also $p(c) \geq p(c_r - h_0/c_r)$, so (43) and (180) in Lemma A.2 give

$$\sup_{u_q \leq c \leq c_r - h_0/c_r} \frac{q(r + \tau_r)}{p(c)} \leq \frac{q(r + \tau_r)}{p(c_r - h_0/c_r)} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p(c_r - h_0/c_r)} \rightarrow (1 - q)e^{-h_0}.$$

Taking the supremum over $z \in K$ and then the limsup in r proves (53).

Proof of (54). By (53), for sufficiently small r ,

$$\sup_{z \in K} \sup_{u_q \leq c \leq c_r - h_0/c_r} \bar{R}_{r,z}(c) < 1.$$

Thus no first crossing can occur at or before $c_r - h_0/c_r$, uniformly for $z \in K$. At the upper point, (52) with $h = h_0$ gives

$$\bar{R}_{r,z} \left(c_r + \frac{h_0}{c_r} \right) = q + (1 - q)e^{h_0} + o(1) > 1.$$

Thus, for sufficiently small r , the first crossing satisfies

$$c_r - \frac{h_0}{c_r} < C_r(z) \leq c_r + \frac{h_0}{c_r}, \quad z \in K. \quad (61)$$

Since $h_0 > 0$ is arbitrary, this yields $C_r(z) = c_r + o(c_r^{-1})$, uniformly on K .

Proof of (55). By Theorem 2.8, it remains to verify regularity as stated in Definition 2.4, with r fixed. Fix $H > 0$ and write

$$G_{r,z}(c) = \frac{R_{0,r,z}\{p(c)\}}{p(c)}.$$

On $|c - c_r| \leq H/c_r$, all non-nulls are rejected, uniformly over $z \in K$, and hence

$$\bar{R}_{r,z}(c) = q\pi_{0,r}G_{r,z}(c) + \frac{q(r + \tau_r)}{p(c)}. \quad (62)$$

Put $x_{\pm} = (c \mp rz)/\sqrt{1-r^2}$. Throughout this paragraph, the window is the local range $|c - c_r| \leq H/c_r$, with $z \in K$. Direct differentiation gives

$$G'_{r,z}(c) = \frac{\phi(c)}{\Phi(c)}G_{r,z}(c) - \frac{\phi(x_+) + \phi(x_-)}{2\sqrt{1-r^2}\Phi(c)}.$$

Uniformly on this window, Lemmas 2.11 and 2.10 give $G_{r,z}(c) = 1 + o(1)$. We also have $c \sim c_r \rightarrow \infty$. Since $rc_r \rightarrow 0$ and $|c - c_r| \leq H/c_r$,

$$x_{\pm}^2 - c^2 = O_K(rc_r) + O(r^2c_r^2) = o(1), \quad \sqrt{1-r^2} = 1 + o(1),$$

uniformly over the same window. Hence

$$\frac{\phi(x_{\pm})}{\phi(c)} = \exp\left\{-\frac{x_{\pm}^2 - c^2}{2}\right\} = 1 + o(1),$$

and therefore

$$\frac{\phi(x_+) + \phi(x_-)}{2\sqrt{1-r^2}\phi(c)} = \frac{1 + o(1) + 1 + o(1)}{2\{1 + o(1)\}} = 1 + o(1).$$

Finally, (171) in Lemma A.2 gives $\phi(c)/\Phi(c) = c\{1 + o(1)\}$, uniformly for $|c - c_r| \leq H/c_r$, and therefore

$$\frac{\phi(c)}{\Phi(c)} \sim c_r.$$

Consequently,

$$\sup_{z \in K} \sup_{|c - c_r| \leq H/c_r} |G'_{r,z}(c)| = o(c_r).$$

On the same window, write $c = c_r + h/c_r$, where $|h| \leq H$. By (43),

$$\frac{q(r + \tau_r)}{p(c)} = (1 - q\pi_{0,r})\frac{p(c_r)}{p(c)},$$

and (180) in Lemma A.2 gives

$$\frac{p(c_r)}{p(c)} = \frac{p(c_r)}{p(c_r + h/c_r)} \rightarrow e^h,$$

uniformly for $|h| \leq H$. Since $e^h \geq e^{-H}$ on this range and $1 - q\pi_{0,r} \rightarrow 1 - q > 0$, for all sufficiently small r ,

$$\inf_{|c - c_r| \leq H/c_r} \frac{q(r + \tau_r)}{p(c)} \geq \frac{1 - q}{2} \cdot \frac{e^{-H}}{2}.$$

Thus we may take $m_H = (1 - q)e^{-H}/4 > 0$. Since

$$\frac{d}{dc} \frac{q(r + \tau_r)}{p(c)} = \frac{q(r + \tau_r)}{p(c)} \frac{\phi(c)}{\Phi(c)},$$

Together with $\phi(c)/\bar{\Phi}(c) \sim c_r$, uniformly on the same window, this gives

$$\inf_{|c-c_r| \leq H/c_r} \frac{d}{dc} \frac{q(r+\tau_r)}{p(c)} \geq \frac{m_H}{2} c_r$$

for all sufficiently small r . By contrast, $q\pi_{0,r}G'_{r,z}(c) = o(c_r)$ uniformly over $z \in K$ and $|c - c_r| \leq H/c_r$. Differentiating (62) therefore shows that

$$\inf_{z \in K} \inf_{|c-c_r| \leq H/c_r} \bar{R}'_{r,z}(c) > 0 \quad (63)$$

for all sufficiently small r . Proposition 2.5 shows that $C_r(z)$ is regular.

Finally, put

$$h_r(z) = c_r \{C_r(z) - c_r\}.$$

By (54), $h_r(z) \rightarrow 0$ uniformly on K . The null limit established in the proof of (52), together with the expression for $D(z)$ in Theorem 2.8, therefore gives

$$D_r(z) = \bar{R}_{0,r,z} \left(c_r + \frac{h_r(z)}{c_r} \right) \rightarrow q$$

uniformly on K . □

2.6.3 The regime $z \in (-y_q, -1)$

Let $y = -z$. For $1 < y < y_q$, put

$$\kappa(y) = a(y)y'\{a(y)\} > 0. \quad (64)$$

The positivity follows from Lemma A.1, since $a(y) > 0$ and $y'\{a(y)\} > 0$ for $1 < y < y_q$.

We first prove a lemma that is useful for the key result in this regime.

Lemma 2.13. *For every compact $I \subset (1, y_q)$ and every $H > 0$, there is an interval U_I whose closure is contained in $(0, a_q)$ and contains $\{a(y) : y \in I\}$. Define*

$$g_r(\tilde{a}) = \tilde{a} + ry(\tilde{a}). \quad (65)$$

For all sufficiently small r , the function g_r is strictly increasing on $[b_r, a_q]$, and in particular on U_I .

For all sufficiently small r , the equation

$$g_r\{\tilde{\alpha}_r(y, h)\} = g_r\{a(y)\} + rh \quad (66)$$

has a unique solution $\tilde{\alpha}_r(y, h) \in U_I$, uniformly for $y \in I$ and $|h| \leq H$, and

$$\tilde{\alpha}_r(y, h) = a(y) + O(r). \quad (67)$$

Conditional on $Z = -y$, a secondary non-null with $a = \tilde{a}$ is rejected at cutoff $a(y)/r + h$ if and only if

$$\tilde{a} \in [\tilde{\alpha}_r(y, h), a_q].$$

Finally,

$$\log \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} = \kappa(y)h + o(1), \quad (68)$$

uniformly for $y \in I$ and $|h| \leq H$.

Proof. Fix a compact $I \subset (1, y_q)$ and $H > 0$ as in the statement. Let

$$A_I = \{a(y) : y \in I\}.$$

Because $I \subset (1, y_q)$ is compact, A_I is a compact subset of $(0, a_q)$. Choose an interval U_I whose closure is contained in $(0, a_q)$ and such that $A_I \subset U_I$. By Lemma A.1, $y(\cdot)$ is continuously differentiable and $y' > 0$ on $(0, a_q)$. Hence

$$g'_r(\tilde{a}) = 1 + ry'(\tilde{a}) > 0, \quad 0 < \tilde{a} < a_q,$$

so g_r is strictly increasing on $[b_r, a_q]$, and in particular on U_I .

Put $d_I = \text{dist}(A_I, U_I^c) > 0$. If $rH < d_I/2$, then $a(y) \pm rH \in U_I$ for all $y \in I$. Since $g'_r \geq 1$ on U_I , the intermediate value theorem gives a unique solution to (66) in U_I , and

$$|\tilde{\alpha}_r(y, h) - a(y)| \leq rH.$$

The mean-value theorem gives, for some ξ_r between $\tilde{\alpha}_r(y, h)$ and $a(y)$,

$$\tilde{\alpha}_r(y, h) - a(y) = \frac{rh}{1 + ry'(\xi_r)}. \quad (69)$$

Since y' is bounded on $\overline{U_I}$ and $|h| \leq H$, this gives $\tilde{\alpha}_r(y, h) = a(y) + O(r)$, uniformly on I . Since y' is uniformly continuous on $\overline{U_I}$ and $|\tilde{\alpha}_r(y, h) - a(y)| \leq rH$, we have $y'(\xi_r) = y'\{a(y)\} + o(1)$ uniformly. Hence

$$\frac{\tilde{\alpha}_r(y, h)}{r} = \frac{a(y)}{r} + h - ry'\{a(y)\}h + o(r), \quad (70)$$

uniformly for $y \in I$ and $|h| \leq H$.

For a secondary non-null with index $\tilde{a} \in [b_r, a_q]$,

$$(\Theta_{\tilde{a}}, B_{\tilde{a}}) = \left(\frac{\tilde{a}}{r} + y(\tilde{a}), 1 \right).$$

Conditional on $Z = -y$,

$$T_{\tilde{a}} = \frac{\tilde{a}}{r} + y(\tilde{a}) - y.$$

Uniformly for $y \in I$ and $\tilde{a} \in [b_r, a_q]$,

$$T_{\tilde{a}} \geq \frac{b_r}{r} + y(b_r) - \sup_{u \in I} u \longrightarrow \infty$$

by (47) in Lemma 2.10. Also $a(y)/r + h > 0$ uniformly for $y \in I$ and $|h| \leq H$. Therefore rejection at cutoff $a(y)/r + h$ is equivalent to

$$\frac{\tilde{a}}{r} + y(\tilde{a}) - y \geq \frac{a(y)}{r} + h,$$

or, equivalently,

$$g_r(\tilde{a}) \geq g_r\{a(y)\} + rh.$$

Since g_r is strictly increasing, a secondary non-null is rejected exactly when its index lies in $[\tilde{\alpha}_r(y, h), a_q]$.

Equations (69) and (70) give, for some $\eta_r(y, h)$ satisfying

$$\sup_{y \in I, |h| \leq H} \frac{|\eta_r(y, h)|}{r} \longrightarrow 0,$$

that

$$\frac{\tilde{\alpha}_r(y, h)}{r} = \frac{a(y)}{r} + h - ry'\{a(y)\}h + \eta_r(y, h).$$

Because $a(y)$ is bounded away from zero on I , $a(y)/r \rightarrow \infty$ uniformly. Applying (179) in Lemma A.2 to the two bounded shifts from $a(y)/r$ gives

$$\log \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} = -\frac{a(y)}{r} \{-ry'\{a(y)\}h + \eta_r(y, h)\} - \frac{(h - ry'\{a(y)\}h + \eta_r(y, h))^2 - h^2}{2} + o(1).$$

The first line equals $\kappa(y)h + o(1)$ uniformly, and the quadratic difference is $o(1)$ uniformly. This proves (68). $\square$

The following lemma identifies the analogue of Lemma 2.12.

Lemma 2.14. *For every compact $I \subset (1, y_q)$ and every $H > 0$, uniformly for $y \in I$ and $|h| \leq H$,*

$$\bar{R}_{r,-y} \left(\frac{a(y)}{r} + h \right) \longrightarrow qM(y) + \{1 - qM(y)\}e^{\kappa(y)h}. \quad (71)$$

Moreover, for every $h_0 > 0$, there is $\gamma > 0$ such that, for all sufficiently small r ,

$$\sup_{y \in I} \sup_{u_q \leq c \leq a(y)/r - h_0} \bar{R}_{r,-y}(c) \leq 1 - \gamma. \quad (72)$$

Consequently, uniformly for $y \in I$,

$$C_r(-y) - \frac{a(y)}{r} \longrightarrow 0 \quad (73)$$

and

$$D_r(-y) \longrightarrow qM(y). \quad (74)$$

Moreover, for sufficiently small r , $C_r(-y)$ is regular for every $y \in I$.

Proof. Proof of (71). Fix $H > 0$, take $y \in I$, and take $|h| \leq H$. By Lemma 2.13, a secondary non-null with $a = \tilde{a}$ is rejected at cutoff $a(y)/r + h$ if and only if

$$\tilde{a} \in [\tilde{\alpha}_r(y, h), a_q].$$

By (41), their mass under the secondary component is

$$\tau_r \mu_r([\tilde{\alpha}_r(y, h), a_q]) = W_r\{\tilde{\alpha}_r(y, h)\}.$$

The definition (39) therefore gives

$$\frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)}.$$

By (67), $y(\tilde{\alpha}_r(y, h)) \rightarrow y$ uniformly for $y \in I$ and $|h| \leq H$. Equation (68) gives

$$\frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} \longrightarrow e^{\kappa(y)h},$$

uniformly for $y \in I$ and $|h| \leq H$. Hence

$$\frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)} \longrightarrow \{1 - qM(y)\}e^{\kappa(y)h} \quad (75)$$

The primary non-null is not rejected at cutoff $a(y)/r + h$. For the index i corresponding to it, $(\Theta_i, B_i) = (c_r + y(b_r), 1)$. Conditional on $Z = -y$,

$$T_i = c_r + y(b_r) - y.$$

Let $a_I = \inf_{u \in I} a(u) > 0$. For $|h| \leq H$,

$$\sup_{y \in I} r|T_i| \leq rc_r + r \sup_{y \in I} |y(b_r) - y| \longrightarrow 0, \quad \inf_{y \in I} r \left\{ \frac{a(y)}{r} + h \right\} \geq a_I - rH \longrightarrow a_I > 0,$$

using (47) and (48) in Lemma 2.10. Hence $|T_i| < a(y)/r + h$ for sufficiently small r , uniformly over $y \in I$ and $|h| \leq H$, so the primary non-null is not rejected at this cutoff.

Since $a(y)$ is bounded away from zero on I and $a(y) \leq a_q$, for all sufficiently small r ,

$$u_q \leq \frac{a(y)}{r} + h \leq \frac{a_q + 1}{r}$$

uniformly for $y \in I$ and $|h| \leq H$. Thus (49) in Lemma 2.11, applied with $A = a_q + 1$ and $T = \sup_{u \in I} u$, gives

$$\bar{R}_{0,r,-y}(a(y)/r + h) \longrightarrow qL_{a(y)}(y) = qM(y). \quad (76)$$

Combining (75) and (76) proves (71).

Proof of (72). Since $I \subset (1, y_q)$ is compact, $y_I := \inf I > 1$. By (47) in Lemma 2.10, $y(b_r) \downarrow 1$. Hence, with

$$\delta := \frac{y_I - 1}{2} > 0,$$

we have $y(b_r) \leq 1 + \delta < y_I$ for sufficiently small r , and therefore

$$y - y(b_r) \geq y_I - y(b_r) \geq \delta, \quad y \in I.$$

Put

$$s_r(y) = c_r + y(b_r) - y.$$

Then

$$\delta \leq y - y(b_r) \leq \sup_{u \in I} u - y(b_r) < \infty, \quad s_r(y) \leq c_r - \delta.$$

Conditional on $Z = -y$, the primary non-null has value $T_i = s_r(y)$, which is positive for sufficiently small r . Hence this primary non-null is rejected at cutoff s exactly when $s \leq s_r(y)$. For $u_q \leq s \leq s_r(y)$, (44) and the bound by one on the secondary integral give

$$\bar{R}_{r,-y}(s) \leq q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)} + \frac{qr}{p(s)} + \frac{q\tau_r}{p(s)}.$$

Since I is compact, $T_I := \sup_{y \in I} y < \infty$. Also $s \leq s_r(y) \leq c_r - \delta$, so (48) gives

$$0 \leq rs \leq rc_r \longrightarrow 0$$

uniformly for $y \in I$ and $u_q \leq s \leq s_r(y)$. Thus, for any fixed $A > 0$, this cutoff range lies in $0 \leq s \leq A/r$ for all sufficiently small r . Applying (49) in Lemma 2.11 with $T = T_I$ gives

$$\frac{R_{0,r,-y}\{p(s)\}}{p(s)} = L_{rs}(y) + o(1)$$

uniformly for $y \in I$ and $u_q \leq s \leq s_r(y)$. Since $rs \rightarrow 0$ uniformly and $y \leq T_I$,

$$L_{rs}(y) = \exp \left\{ -\frac{(rs)^2}{2} \right\} \cosh(rsy) = 1 + o(1)$$

uniformly for $y \in I$ and $u_q \leq s \leq s_r(y)$. Since $\pi_{0,r} \rightarrow 1$, it follows that

$$q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)} \leq q + o(1).$$

Also p is decreasing, so $p(s) \geq p(s_r(y))$. Therefore, for any such cutoff,

$$\bar{R}_{r,-y}(s) \leq q + o(1) + \frac{q(r + \tau_r)}{p(s_r(y))}$$

uniformly for $y \in I$. By (43),

$$\frac{q(r + \tau_r)}{p(s_r(y))} = (1 - q\pi_{0,r}) \frac{p(c_r)}{p\{c_r + y(b_r) - y\}}.$$

Since $c_r \rightarrow \infty$ by (48) and

$$\delta \leq y - y(b_r) \leq \sup_{u \in I} u - 1 =: \Delta_I^{\text{shift}} < \infty, \quad (77)$$

we have $s_r(y) = c_r + y(b_r) - y \rightarrow \infty$, uniformly for $y \in I$. Equation (77) shows that the shift $y - y(b_r)$ lies in the fixed compact interval $[\delta, \Delta_I^{\text{shift}}] \subset (0, \infty)$, so (181) in Lemma A.2 gives

$$\frac{p(c_r)}{p\{c_r + y(b_r) - y\}} = \frac{p\{s_r(y) + y - y(b_r)\}}{p(s_r(y))} \rightarrow 0$$

uniformly for $y \in I$. Hence

$$\sup_{y \in I} \sup_{u_q \leq s \leq s_r(y)} \bar{R}_{r,-y}(s) \leq q + o(1) < 1. \quad (78)$$

Fix $h_0 > 0$. The complementary range $s_r(y) < s \leq a(y)/r - h_0$ is nonempty uniformly on I , because

$$\inf_{y \in I} \left\{ \frac{a(y)}{r} - h_0 - s_r(y) \right\} \geq \frac{\inf_{y \in I} a(y)}{r} - c_r - O_I(1) \rightarrow \infty,$$

where $rc_r \rightarrow 0$ by (48) in Lemma 2.10. It remains to prove that there is $\gamma > 0$ such that, for all sufficiently small r ,

$$\sup_{y \in I} \sup_{s_r(y) < s \leq a(y)/r - h_0} \bar{R}_{r,-y}(s) \leq 1 - \gamma \quad (79)$$

For $s > s_r(y)$, the identity $T_i = s_r(y) > 0$ gives $|T_i| = T_i < s$. For sufficiently small r , every secondary non-null value is positive uniformly over $y \in I$ and $\tilde{a} \in [b_r, a_q]$. On the range $s_r(y) < s \leq a(y)/r - h_0$, define

$$\beta_{r,y,s} := \inf \left\{ \tilde{a} \in [b_r, a_q] : \frac{\tilde{a}}{r} + y(\tilde{a}) - y \geq s \right\}. \quad (80)$$

The set is nonempty for sufficiently small r , since $\tilde{a} = a_q$ satisfies the inequality uniformly over $y \in I$. The map $\tilde{a} \mapsto \tilde{a}/r + y(\tilde{a}) - y$ is strictly increasing by Lemma A.1, so a secondary non-null with $a = \tilde{a}$ is rejected if and only if $\tilde{a} \in [\beta_{r,y,s}, a_q]$. Therefore, by (44),

$$\bar{R}_{r,-y}(s) = q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)} + \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]). \quad (81)$$

Suppose (79) fails for this h_0 . Then there are

$$r_n \downarrow 0, \quad y_n \in I, \quad s_{r_n}(y_n) < s_n \leq a(y_n)/r_n - h_0, \quad \liminf_{n \rightarrow \infty} \bar{R}_{r_n,-y_n}(s_n) \geq 1. \quad (82)$$

After taking a subsequence,

$$y_n \rightarrow y \in I, \quad a(y_n) \rightarrow a(y), \quad x_n = r_n s_n \rightarrow x \in [0, a(y)]. \quad (83)$$

Define

$$h_n := s_n - \frac{a(y_n)}{r_n}.$$

Then $h_n \leq -h_0$ by (82). If $h_n = O(1)$, then after passing to a further subsequence $h_n \rightarrow h \leq -h_0$. Equation (71) gives

$$\bar{R}_{r_n, -y_n}(s_n) \rightarrow qM(y) + \{1 - qM(y)\}e^{\kappa(y)h} \leq qM(y) + \{1 - qM(y)\}e^{-\kappa(y)h_0} < 1,$$

contradicting $\liminf_{n \rightarrow \infty} \bar{R}_{r_n, -y_n}(s_n) \geq 1$. Hence, after passing to a further subsequence if necessary, it remains to consider

$$h_n \rightarrow -\infty. \quad (84)$$

We next identify the rejected secondary non-nulls at cutoff s_n . A secondary non-null with index $\tilde{a} \in [b_{r_n}, a_q]$ has $(\Theta, B) = (\tilde{a}/r_n + y(\tilde{a}), 1)$. Conditional on $Z = -y_n$, its value is

$$T_{\tilde{a}, n} = \frac{\tilde{a}}{r_n} + y(\tilde{a}) - y_n.$$

For all large n , this value is positive uniformly in $\tilde{a} \in [b_{r_n}, a_q]$, because $b_{r_n}/r_n \rightarrow \infty$ and $y(\tilde{a}) - y_n$ is bounded below. Hence $|T_{\tilde{a}, n}| = T_{\tilde{a}, n}$, and the rejection rule $|T_{\tilde{a}, n}| \geq s_n$ is equivalent to

$$\frac{\tilde{a}}{r_n} + y(\tilde{a}) - y_n \geq s_n.$$

The rejected set is nonempty because the endpoint $\tilde{a} = a_q$ is rejected. Indeed, compactness of $I \subset (1, y_q)$ and continuity of $a(\cdot)$ give

$$\Delta_I := a_q - \sup_{u \in I} a(u) > 0.$$

Since $s_n \leq a(y_n)/r_n - h_0$ by (82),

$$\frac{a_q}{r_n} + y_q - y_n - s_n \geq \frac{a_q - a(y_n)}{r_n} + y_q - y_n + h_0 > 0$$

for large n , because $a_q - a(y_n) \geq \Delta_I$ and $y_n \leq \sup I < y_q$. Define

$$G_n(\tilde{a}) := g_{r_n}(\tilde{a}) - r_n(s_n + y_n), \quad \tilde{a} \in [b_{r_n}, a_q],$$

where g_r is defined in (65). Multiplying the rejection inequality by r_n , a secondary non-null with index $\tilde{a}$ is rejected if and only if

$$G_n(\tilde{a}) \geq 0.$$

By the monotonicity part of Lemma 2.13, G_n is continuous and strictly increasing on $[b_{r_n}, a_q]$ for all large n . Therefore a secondary non-null with $a = \tilde{a}$ is rejected if and only if $\tilde{a} \in [\beta_n, a_q]$, where

$$\beta_n := \inf \{ \tilde{a} \in [b_{r_n}, a_q] : G_n(\tilde{a}) \geq 0 \}. \quad (85)$$

Put

$$d_n := y_n - y(\beta_n). \quad (86)$$

If $\beta_n > b_{r_n}$, then $G_n(\beta_n) = 0$. Thus, in this case,

$$\frac{\beta_n}{r_n} - s_n = d_n. \quad (87)$$

If $\beta_n = b_{r_n}$, then $G_n(\beta_n) \geq 0$, so (87) is replaced by

$$\frac{\beta_n}{r_n} - s_n \geq d_n. \quad (88)$$

Moreover, $s_n \rightarrow \infty$, because $s_n > s_{r_n}(y_n) = c_{r_n} + y(b_{r_n}) - y_n$, $c_{r_n} \rightarrow \infty$, $y(b_{r_n}) \rightarrow 1$, and I is compact. Also $d_n > 0$ eventually. If $\beta_n = b_{r_n}$, this follows from the definition of d_n and (77), which give

$$d_n = y_n - y(b_{r_n}) \geq \delta > 0.$$

If $\beta_n > b_{r_n}$, then

$$g_{r_n}(\beta_n) = r_n(s_n + y_n) = g_{r_n}\{a(y_n)\} + r_n h_n \leq g_{r_n}\{a(y_n)\} - r_n h_0.$$

The strict monotonicity of g_{r_n} therefore gives $\beta_n < a(y_n)$, and hence $d_n = y_n - y(\beta_n) > 0$. The null term in (81) already has a limit along this subsequence:

$$q\pi_{0,r_n} \frac{R_{0,r_n,-y_n}\{p(s_n)\}}{p(s_n)} \longrightarrow qL_x(y) \leq qM(y) < 1. \quad (89)$$

Indeed, (49) in Lemma 2.11 applies because $r_n s_n \rightarrow x$, $y_n \rightarrow y$, and $\pi_{0,r_n} \rightarrow 1$. The inequality follows from $M(y) = \sup_{\alpha \geq 0} L_\alpha(y)$, the strict monotonicity of M on $(1, \infty)$, $y \in I \subset (1, y_q)$, and $qM(y_q) = 1$. Equations (81) and (89), together with the threshold characterization in (85), imply

$$\bar{R}_{r_n,-y_n}(s_n) = qL_x(y) + o(1) + \frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]). \quad (90)$$

Moreover, (41), (39), (167) in Lemma A.1, and the bound from (87)–(88) give

$$\frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]) = \frac{qW_{r_n}(\beta_n)}{p(s_n)} = [1 - qM\{y(\beta_n)\}] \frac{p(\beta_n/r_n)}{p(s_n)} \leq \frac{p(s_n + d_n)}{p(s_n)}. \quad (91)$$

It remains only to prove that

$$s_n d_n \longrightarrow \infty. \quad (92)$$

Equations (91), (92), and (182) in Lemma A.2 imply

$$\frac{q\tau_{r_n}}{p(s_n)}\mu_{r_n}([\beta_n, a_q]) = o(1).$$

Substituting this into (90) gives

$$\bar{R}_{r_n,-y_n}(s_n) \longrightarrow qL_x(y) < 1,$$

which yields a contradiction to (82).

To prove (92), we now consider three cases under (82), (83), and (84).

Case (a): $0 < x < a(y)$. Suppose $x > 0$. Since $b_{r_n} \rightarrow 0$ by (47) in Lemma 2.10 and $r_n s_n \rightarrow x$, the endpoint case $\beta_n = b_{r_n}$ cannot occur for large n : multiplying (88) by r_n would give

$$b_{r_n} - r_n s_n \geq r_n \{y_n - y(b_{r_n})\},$$

whose left side tends to $-x < 0$, while the right side is nonnegative for large n . Hence (87) holds eventually. Multiplying it by r_n gives

$$\beta_n - r_n s_n = r_n d_n. \quad (93)$$

The factor d_n is bounded. Indeed, I is compact, so $|y_n| \leq \sup_{u \in I} |u|$. Also $\beta_n \in [b_{r_n}, a_q]$, and $y(\cdot)$ is increasing by Lemma A.1; hence

$$y(b_{r_n}) \leq y(\beta_n) \leq y(a_q) = y_q. \quad (94)$$

Since $y(b_{r_n}) \downarrow 1$ by (47) in Lemma 2.10, the sequence $y(\beta_n)$ is bounded. Thus $r_n d_n \rightarrow 0$, so $\beta_n - r_n s_n \rightarrow 0$. Since $r_n s_n \rightarrow x$, we have $\beta_n \rightarrow x$. Since $x < a(y)$ and $y(\cdot)$ is strictly increasing by Lemma A.1, $y(x) < y$.

Together, $y_n \rightarrow y$ from (83) and $\beta_n \rightarrow x$ imply, by continuity of $y(\cdot)$, that $d_n \rightarrow d := y - y(x) > 0$. Also, (87) gives $\beta_n/r_n = s_n + d_n$. Therefore, for all large n ,

$$\frac{\beta_n}{r_n} = s_n + d_n, \quad 0 < \frac{d}{2} \leq d_n \leq 2d, \quad s_n \rightarrow \infty.$$

As a result, (92) is proved.

Case (b): $x = 0$. Then $r_n s_n \rightarrow 0$. Since

$$s_n > s_{r_n}(y_n) = c_{r_n} + y(b_{r_n}) - y_n,$$

(48) gives $c_{r_n} \rightarrow \infty$, while (47) gives $y(b_{r_n}) \downarrow 1$ and compactness of I keeps y_n bounded. Hence $s_n \rightarrow \infty$. If $\beta_n = b_{r_n}$, then $\beta_n \rightarrow 0$ by (47). If $\beta_n > b_{r_n}$, then (87) holds; multiplying it by r_n , using compactness of I , and applying (94) gives

$$\beta_n - r_n s_n = r_n d_n \rightarrow 0.$$

Since $r_n s_n \rightarrow 0$, this also gives $\beta_n \rightarrow 0$. Thus $\beta_n \rightarrow 0$ in both cases. By (83), $y_n \rightarrow y$, and by the inverse relation together with (165) in Lemma A.1, $y(\beta_n) \rightarrow 1$. Hence

$$d_n \rightarrow y - 1 > 0.$$

Moreover, (87) in the interior case and (88) in the endpoint case give $\beta_n/r_n \geq s_n + d_n$. Since $d_n \rightarrow d := y - 1 > 0$ and $s_n \rightarrow \infty$, (92) is proved.

Case (c): $x = a(y)$. By (84), this is the only remaining subcase. Since $r_n s_n \rightarrow a(y)$ and $a(y_n) \rightarrow a(y)$, we have

$$r_n h_n = r_n s_n - a(y_n) \rightarrow a(y) - a(y) = 0.$$

The threshold equation $G_n(\tilde{a}) = 0$ is

$$g_{r_n}(\tilde{a}) = r_n(s_n + y_n) = a(y_n) + r_n y_n + r_n h_n = g_{r_n}\{a(y_n)\} + r_n h_n,$$

where $y\{a(y_n)\} = y_n$. Since $h_n < 0$ and g_{r_n} is strictly increasing by Lemma 2.13, the corresponding threshold lies below $a(y_n)$. To justify the existence and location of this threshold, choose $\eta > 0$ such that

$$[a(y_n) - \eta, a(y_n)] \subset (0, a_q)$$

for all large n ; this is possible because $a(y_n) \rightarrow a(y) \in (0, a_q)$. Since $-r_n h_n \rightarrow 0$ and $g'_{r_n} \geq 1$, for all large n ,

$$g_{r_n}\{a(y_n) - \eta\} < g_{r_n}\{a(y_n)\} + r_n h_n < g_{r_n}\{a(y_n)\}.$$

Continuity and strict monotonicity of g_{r_n} therefore give a unique $\tilde{\beta}_n \in (a(y_n) - \eta, a(y_n))$ satisfying

$$g_{r_n}(\tilde{\beta}_n) = g_{r_n}\{a(y_n)\} + r_n h_n.$$

Moreover, the mean-value theorem and $g'_{r_n} \geq 1$ give

$$0 < a(y_n) - \tilde{\beta}_n \leq -r_n h_n \rightarrow 0.$$

Because $a(y_n)$ remains bounded away from 0 and a_q , and $b_{r_n} \rightarrow 0$ by (47) in Lemma 2.10, it follows that

$$b_{r_n} < \tilde{\beta}_n < a_q$$

eventually. Thus $\tilde{\beta}_n$ is the threshold β_n in (85), so $G_n(\beta_n) = 0$. The mean-value theorem applied to g_{r_n} now gives, for some ξ_n between β_n and $a(y_n)$,

$$-r_n h_n = g_{r_n}\{a(y_n)\} - g_{r_n}(\beta_n) = (a(y_n) - \beta_n)\{1 + r_n y'(\xi_n)\}.$$

Therefore $\beta_n < a(y_n)$ and

$$a(y_n) - \beta_n = \frac{-r_n h_n}{1 + r_n y'(\xi_n)}. \quad (95)$$

For some ζ_n between β_n and $a(y_n)$,

$$\frac{\beta_n}{r_n} - s_n = d_n = y'(\zeta_n)\{a(y_n) - \beta_n\}. \quad (96)$$

Combining (95) and (96) gives

$$s_n d_n = (r_n s_n) \frac{y'(\zeta_n)}{1 + r_n y'(\xi_n)} (-h_n). \quad (97)$$

Here $r_n s_n \rightarrow a(y) > 0$. Also $\beta_n \rightarrow a(y)$ and $a(y_n) \rightarrow a(y)$, so $\xi_n, \zeta_n \rightarrow a(y)$; hence $y'(\zeta_n) \rightarrow y'\{a(y)\} > 0$, while $1 + r_n y'(\xi_n) \rightarrow 1$. Since $h_n \rightarrow -\infty$ by (84), (97) yields

$$s_n d_n \rightarrow \infty.$$

Thus (92) is proved.

This proves (79). Combining (78) and (79), and reducing γ if necessary, proves (72).

Proof of (73). Fix $h > 0$. Equation (72) excludes a crossing before $a(y)/r - h$. Also,

$$qM(y) + \{1 - qM(y)\}e^{\kappa(y)h} > 1, \quad y \in I,$$

and the left side is bounded away from one on I , since I is compact, $qM(y) < 1$, and $\kappa(y) > 0$. Therefore (71), applied at h , gives a strict crossing by $a(y)/r + h$, uniformly on I . Hence

$$\frac{a(y)}{r} - h < C_r(-y) \leq \frac{a(y)}{r} + h.$$

Let $h \downarrow 0$. Then

$$C_r(-y) - \frac{a(y)}{r} \rightarrow 0, \quad rC_r(-y) \rightarrow a(y),$$

uniformly on I .

Proof of (74). We now verify regularity with r fixed. Fix $H = 1$, and, for $y \in I$ and $|h| \leq H$, let $\tilde{\alpha}_r(y, h)$ be the threshold defined by (66). Write

$$\mathcal{N}_r(y, h) = q\pi_{0,r} \frac{R_{0,r,-y}\{p(a(y)/r + h)\}}{p(a(y)/r + h)},$$

$$\mathcal{S}_r(y, h) = \frac{qW_r\{\tilde{\alpha}_r(y, h)\}}{p(a(y)/r + h)} = [1 - qM\{y(\tilde{\alpha}_r(y, h))\}] \frac{p\{\tilde{\alpha}_r(y, h)/r\}}{p(a(y)/r + h)}.$$

The primary non-null is not rejected on this window. Indeed, for the corresponding index i , conditional on $Z = -y$,

$$T_i = c_r + y(b_r) - y.$$

Uniformly for $y \in I$ and $|h| \leq H$,

$$r|T_i| \leq rc_r + r \sup_{u \in I} |y(b_r) - u| \rightarrow 0,$$

whereas

$$r \left\{ \frac{a(y)}{r} + h \right\} = a(y) + rh \geq \inf_{u \in I} a(u) - rH$$

is bounded away from zero for all sufficiently small r . Hence $|T_i| < a(y)/r + h$ throughout the window, so the primary non-null is not rejected. By Lemma 2.13,

$$\bar{R}_{r,-y} \left(\frac{a(y)}{r} + h \right) = \mathcal{N}_r(y, h) + \mathcal{S}_r(y, h).$$

We first differentiate the secondary non-null contribution $\mathcal{S}_r(y, h)$. Equation (66) gives

$$\frac{\partial}{\partial h} \tilde{\alpha}_r(y, h) = \frac{r}{1 + ry' \{ \tilde{\alpha}_r(y, h) \}}.$$

Put $\lambda(x) = \phi(x)/\bar{\Phi}(x)$. Since $-p'(x)/p(x) = \lambda(x)$, direct differentiation gives

$$\begin{aligned} \frac{\partial}{\partial h} \log \mathcal{S}_r(y, h) = & - \frac{qM' \{ y(\tilde{\alpha}_r(y, h)) \} y' \{ \tilde{\alpha}_r(y, h) \}}{1 - qM \{ y(\tilde{\alpha}_r(y, h)) \}} \frac{r}{1 + ry' \{ \tilde{\alpha}_r(y, h) \}} \\ & - \frac{\lambda \{ \tilde{\alpha}_r(y, h)/r \}}{1 + ry' \{ \tilde{\alpha}_r(y, h) \}} + \lambda \left(\frac{a(y)}{r} + h \right). \end{aligned}$$

The first term is $O(r)$ uniformly, because I is compact and $1 - qM \{ y(\tilde{\alpha}_r(y, h)) \}$ is bounded away from zero by (67) and (167) in Lemma A.1. Equations (67) and (70), together with (171) in Lemma A.2, give

$$\lambda \left(\frac{a(y)}{r} + h \right) - \frac{\lambda \{ \tilde{\alpha}_r(y, h)/r \}}{1 + ry' \{ \tilde{\alpha}_r(y, h) \}} = a(y)y' \{ a(y) \} + o(1) = \kappa(y) + o(1)$$

uniformly for $y \in I$ and $|h| \leq H$. Combining this with (75) yields

$$\frac{\partial}{\partial h} \mathcal{S}_r(y, h) \longrightarrow \kappa(y) \{ 1 - qM(y) \} e^{\kappa(y)h} \quad (98)$$

uniformly on the same set.

It remains to differentiate the null contribution. Put

$$x_{\pm} = \frac{a(y)/r + h \pm ry}{\sqrt{1 - r^2}}, \quad Q_{\pm} = \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(a(y)/r + h)}.$$

Then

$$\frac{R_{0,r,-y} \{ p(a(y)/r + h) \}}{p(a(y)/r + h)} = \frac{Q_+ + Q_-}{2},$$

and

$$\frac{\partial Q_{\pm}}{\partial h} = Q_{\pm} \left\{ \lambda \left(\frac{a(y)}{r} + h \right) - \frac{\lambda(x_{\pm})}{\sqrt{1 - r^2}} \right\}.$$

Since $a(y)$ is bounded away from zero on I and $a(y) \leq a_q$, for all sufficiently small r ,

$$u_q \leq \frac{a(y)}{r} + h \leq \frac{a_q + 1}{r}$$

uniformly for $y \in I$ and $|h| \leq H$. Applying (49) in Lemma 2.11 with $A = a_q + 1$ and $T = \sup_{u \in I} u$ therefore shows that the quantities $Q_{\pm}$ are uniformly bounded. Moreover, (171) in Lemma A.2 gives

$$\lambda \left(\frac{a(y)}{r} + h \right) - \frac{\lambda(x_{\pm})}{\sqrt{1 - r^2}} = \left(\frac{a(y)}{r} + h \right) - \frac{a(y)/r + h \pm ry}{1 - r^2} + O(r) = O(r)$$

uniformly. Therefore

$$\sup_{y \in I} \sup_{|h| \leq H} \left| \frac{\partial}{\partial h} \mathcal{N}_r(y, h) \right| \longrightarrow 0. \quad (99)$$

Since I is compact and $\kappa(y)\{1 - qM(y)\}e^{\kappa(y)h}$ is strictly positive on $I \times [-H, H]$, (98) and (99) imply

$$\inf_{y \in I} \inf_{|h| \leq H} \frac{\partial}{\partial h} \bar{R}_{r,-y} \left(\frac{a(y)}{r} + h \right) > 0$$

for all sufficiently small r . Proposition 2.5 shows that $C_r(-y)$ is regular.

Finally, put

$$h_r(y) = C_r(-y) - \frac{a(y)}{r}.$$

By (73), $h_r(y) \rightarrow 0$ uniformly on I . The null limit (76), together with the expression for $D(z)$ in Theorem 2.8, therefore gives

$$D_r(-y) = \bar{R}_{0,r,-y} \left(\frac{a(y)}{r} + h_r(y) \right) \longrightarrow qM(y)$$

uniformly on I . □

2.6.4 The regime $z < -y_q$

Let $y = -z$. For $y > y_q$, let $\alpha_q(y) \in (0, a_q)$ be the unique solution of

$$qL_{\alpha_q(y)}(y) = 1. \tag{100}$$

For fixed y , the map $a \mapsto L_a(y)$ is continuous and

$$\frac{\partial}{\partial a} \log L_a(y) = -a + y \tanh(ay). \tag{101}$$

By Lemma A.1, this derivative is positive on $(0, a(y))$. Since $y > y_q$, the same lemma gives $a(y) > a_q$, so $a \mapsto L_a(y)$ is strictly increasing on $[0, a_q]$. Moreover, by the definition of a_q in (36),

$$qL_0(y) = q < 1, \quad qL_{a_q}(y) > qL_{a_q}(y_q) = qM(y_q) = 1,$$

where the strict inequality follows from $a_q > 0$, $y > y_q$, and the monotonicity of $\cosh(a_q y)$ in $y > 0$. Thus continuity and strict monotonicity give exactly one solution to (100).

The next lemma analyzes the limiting conditional cutoff and FDP.

Lemma 2.15. *For every compact $I \subset (y_q, \infty)$, uniformly for $y \in I$,*

$$rC_r(-y) \longrightarrow \alpha_q(y) \tag{102}$$

and

$$D_r(-y) \longrightarrow 1. \tag{103}$$

Proof. Proof of (102). Fix a compact $I \subset (y_q, \infty)$, and put

$$\delta_I = \inf_{y \in I} (y - y_q) > 0.$$

Let

$$s_r(y) = c_r + y(b_r) - y.$$

Recall from Lemma 2.13 that, for $\tilde{a} \in [b_r, a_q]$, $g_r(\tilde{a}) = \tilde{a} + ry'(\tilde{a})$. By Lemma A.1, $g'_r(\tilde{a}) = 1 + ry'(\tilde{a}) > 0$. For $s > s_r(y)$, define

$$\mathcal{A}_{r,y,s} := \{\tilde{a} \in [b_r, a_q] : g_r(\tilde{a}) \geq r(s + y)\}, \quad \beta_{r,y,s} := \begin{cases} \inf \mathcal{A}_{r,y,s}, & \mathcal{A}_{r,y,s} \neq \emptyset, \\ a_q, & \mathcal{A}_{r,y,s} = \emptyset. \end{cases} \tag{104}$$

For all sufficiently small r , the primary non-null is not rejected when $s > s_r(y)$. To verify the corresponding sign assertion for the secondary non-nulls, note that, conditional on $Z = -y$, the value associated with $\tilde{a} \in [b_r, a_q]$ is

$$\frac{\tilde{a}}{r} + y(\tilde{a}) - y \geq \frac{b_r}{r} + y(b_r) - \sup_{u \in I} u. \quad (105)$$

The monotonicity of $y(\cdot)$ gives (105). Its right-hand side tends to infinity as $r \downarrow 0$, by (47) in Lemma 2.10. Hence every secondary non-null value is positive for all sufficiently small r , uniformly over $y \in I$ and $\tilde{a} \in [b_r, a_q]$. Thus the rejected secondary non-nulls are indexed by $\mathcal{A}_{r,y,s}$. If this set is nonempty, continuity and strict monotonicity of g_r show that it is the closed interval $[\beta_{r,y,s}, a_q]$ and that

$$g_r(\beta_{r,y,s}) \geq r(s + y).$$

Consequently, in the nonempty case,

$$\frac{\beta_{r,y,s}}{r} - s \geq y - y(\beta_{r,y,s}) \geq y - y_q \geq \delta_I. \quad (106)$$

If $\mathcal{A}_{r,y,s} = \emptyset$, its μ_r -mass is zero. Since $\mu_r(\{a_q\}) = 0$ by (41), the convention in (104) gives in both cases

$$\mu_r(\mathcal{A}_{r,y,s}) = \mu_r([\beta_{r,y,s}, a_q]).$$

Using (41), (167) in Lemma A.1, and (106) when the set is nonempty, we obtain

$$0 \leq \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]) \leq \frac{p(s + \delta_I)}{p(s)}.$$

Moreover,

$$\inf_{y \in I} s_r(y) = c_r + y(b_r) - \sup_{y \in I} y \longrightarrow \infty,$$

because $c_r \rightarrow \infty$, $y(b_r) \rightarrow 1$, and I is compact. Therefore (181) in Lemma A.2 gives

$$\sup_{y \in I} \sup_{s > s_r(y)} \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]) \longrightarrow 0. \quad (107)$$

The function $(a, y) \mapsto L_a(y)$ is C^∞ . Equation (101), together with $\alpha_q(y) < a_q < a(y)$ and Lemma A.1, shows that

$$\left. \frac{\partial}{\partial a} \log L_a(y) \right|_{a=\alpha_q(y)} > 0.$$

The implicit function theorem therefore shows that $y \mapsto \alpha_q(y)$ is continuous on (y_q, ∞) . Since I is compact,

$$m_I := \inf_{y \in I} \min\{\alpha_q(y), a_q - \alpha_q(y), a(y) - \alpha_q(y)\} > 0. \quad (108)$$

Fix any $0 < \delta < m_I/2$. For every $y \in I$, continuity and strict monotonicity in a give

$$qL_{\alpha_q(y)-\delta}(y) < qL_{\alpha_q(y)}(y) = 1 < qL_{\alpha_q(y)+\delta}(y).$$

Both strict gaps are continuous functions of y . Their minima over the compact set I are positive, so there is $\gamma > 0$ such that, uniformly on I ,

$$qL_{\alpha_q(y)-\delta}(y) \leq 1 - 2\gamma, \quad qL_{\alpha_q(y)+\delta}(y) \geq 1 + 2\gamma \quad (109)$$

The choice $\delta < m_I/2$ also gives

$$\alpha_q(y) + \delta \leq \alpha_q(y) + \frac{m_I}{2} \leq a(y) - \frac{m_I}{2} < a(y), \quad (110)$$

uniformly for $y \in I$.

Write $g_y(x) = \log L_x(y)$. The proof of Lemma A.1, through (168), shows that $g'_y(x) > 0$ for $0 < x < a(y)$. By (110), every $x \in (0, \alpha_q(y) + \delta)$ lies in $(0, a(y))$. Hence $\partial_x L_x(y) = L_x(y)g'_y(x) > 0$ throughout $(0, \alpha_q(y) + \delta)$, and therefore $x \mapsto L_x(y)$ is increasing on $[0, \alpha_q(y) + \delta]$.

For $u_q \leq s \leq s_r(y)$, the population curve (44) and the fact that the primary and secondary non-null masses sum to $r + \tau_r$ give

$$\bar{R}_{r,-y}(s) \leq q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)} + \frac{q(r + \tau_r)}{p(s)}.$$

Since $s_r(y) = c_r + O_I(1)$,

$$0 \leq rs \leq rc_r + O_I(r) \rightarrow 0$$

uniformly for $y \in I$ and $u_q \leq s \leq s_r(y)$. The set I is bounded, and this cutoff range lies in $[0, A/r]$ for some fixed $A > 0$ when r is sufficiently small. Hence (49) in Lemma 2.11 yields

$$\frac{R_{0,r,-y}\{p(s)\}}{p(s)} = L_{rs}(y) + o(1) = 1 + o(1)$$

uniformly for $y \in I$ and $u_q \leq s \leq s_r(y)$. Since $\pi_{0,r} \rightarrow 1$ and p is decreasing, it follows that

$$\bar{R}_{r,-y}(s) \leq q + o(1) + \frac{q(r + \tau_r)}{p(s_r(y))}.$$

For the remaining term, (43) and $s_r(y) = c_r + y(b_r) - y$ give

$$\frac{q(r + \tau_r)}{p(s_r(y))} = (1 - q\pi_{0,r}) \frac{p\{s_r(y) + y - y(b_r)\}}{p(s_r(y))}.$$

Here $1 - q\pi_{0,r}$ is bounded. Also $s_r(y) \rightarrow \infty$ uniformly on I , and, because $I \subset (y_q, \infty)$ is compact while $y(b_r) \rightarrow 1$, the shifts $y - y(b_r)$ stay in a compact subinterval of $(0, \infty)$. Thus (181) in Lemma A.2 makes the ratio tend to zero uniformly on I . Thus,

$$\sup_{u_q \leq s \leq s_r(y)} \bar{R}_{r,-y}(s) \leq q + o(1), \quad \text{uniformly for } y \in I.$$

For $s > s_r(y)$, the term in (44) that corresponds to the primary non-nulls is zero, and (107) and (49) in Lemma 2.11 give, uniformly for $y \in I$ and $s_r(y) < s \leq (\alpha_q(y) + \delta)/r$,

$$\bar{R}_{r,-y}(s) = q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)} + \frac{q\tau_r}{p(s)} \mu_r([\beta_{r,y,s}, a_q]) = qL_{rs}(y) + o(1).$$

Combining these two ranges gives

$$\sup_{u_q \leq s \leq (\alpha_q(y) + \delta)/r} \bar{R}_{r,-y}(s) \leq qL_{\alpha_q(y) - \delta}(y) + o(1) \leq 1 - \gamma. \quad (111)$$

Since $rs_r(y) \rightarrow 0$ uniformly on I and $\alpha_q(y) + \delta$ is bounded away from zero, we have

$$\frac{\alpha_q(y) + \delta}{r} > s_r(y)$$

for all sufficiently small r , uniformly on I . At the cutoff $(\alpha_q(y) + \delta)/r$, the term in (44) that corresponds to the primary non-nulls is zero, while the secondary term is nonnegative. Therefore (49) in Lemma 2.11 gives, uniformly for $y \in I$,

$$\bar{R}_{r,-y} \left(\frac{\alpha_q(y) + \delta}{r} \right) \geq q\pi_{0,r} \frac{R_{0,r,-y}\{p((\alpha_q(y) + \delta)/r)\}}{p((\alpha_q(y) + \delta)/r)} = qL_{\alpha_q(y) + \delta}(y) + o(1). \quad (112)$$

By (109), the right side is at least $1 + \gamma$ for all sufficiently small r . Hence

$$\alpha_q(y) - \delta < rC_r(-y) \leq \alpha_q(y) + \delta.$$

Letting $\delta \downarrow 0$ gives

$$rC_r(-y) \longrightarrow \alpha_q(y),$$

uniformly for $y \in I$.

We now verify regularity for each fixed sufficiently small r . Fix one $0 < \delta < m_I/2$ as in (108) in Lemma 2.15 and consider the interval

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r}.$$

The primary non-null has conditional value

$$T_i = c_r + y(b_r) - y = s_r(y).$$

It is positive for sufficiently small r , uniformly in $y \in I$, and it is rejected at cutoff s exactly when $s \leq s_r(y)$. Since $\delta < m_I/2$ and $m_I \leq \alpha_q(y)$, we have $\alpha_q(y) - \delta \geq m_I/2$. On the other hand,

$$rs_r(y) = rc_r + r\{y(b_r) - y\} \longrightarrow 0$$

uniformly for $y \in I$, by (48), $y(b_r) \rightarrow 1$, and compactness of I . Therefore, for all sufficiently small r , for every $y \in I$ and every cutoff s satisfying

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r},$$

we have

$$s_r(y) < \frac{\alpha_q(y) - \delta}{r} \leq s.$$

Thus every cutoff in this interval lies strictly to the right of the primary value $s_r(y)$, so the primary non-null is not rejected there. Moreover, uniformly for these y and s ,

$$\begin{aligned} r(s + y) - g_r(b_r) &\geq \alpha_q(y) - \delta + r\{y - y(b_r)\} - b_r \geq \frac{m_I}{2} - o(1) > 0, \\ g_r(a_q) - r(s + y) &\geq a_q - \alpha_q(y) - \delta + r(y_q - y) \geq \frac{m_I}{2} - o(1) > 0. \end{aligned}$$

Strict monotonicity of g_r therefore implies that $\mathcal{A}_{r,y,s} \neq \emptyset$, that $\beta_{r,y,s} \in (b_r, a_q)$, and that

$$g_r(\beta_{r,y,s}) = r(s + y), \quad \frac{\partial \beta_{r,y,s}}{\partial s} = \frac{r}{1 + ry'(\beta_{r,y,s})}.$$

The threshold equation also gives

$$\beta_{r,y,s} = rs + r\{y - y(\beta_{r,y,s})\} = rs + O(r)$$

uniformly. Hence, for all sufficiently small r ,

$$\frac{m_I}{4} \leq \beta_{r,y,s} \leq a_q - \frac{m_I}{4},$$

so the thresholds remain in a fixed compact subinterval of $(0, a_q)$.

For $y \in I$ and cutoffs s in the local crossing interval

$$\frac{\alpha_q(y) - \delta}{r} \leq s \leq \frac{\alpha_q(y) + \delta}{r},$$

write the null and secondary non-null contributions as

$$\mathcal{N}_{r,y}(s) = q\pi_{0,r} \frac{R_{0,r,-y}\{p(s)\}}{p(s)}, \quad \mathcal{S}_{r,y}(s) = [1 - qM\{y(\beta_{r,y,s})\}] \frac{p(\beta_{r,y,s}/r)}{p(s)}.$$

Then $\bar{R}_{r,-y}(s) = \mathcal{N}_{r,y}(s) + \mathcal{S}_{r,y}(s)$. Put $\lambda(x) = \phi(x)/\bar{\Phi}(x)$, and abbreviate $\beta = \beta_{r,y,s}$. Since

$$\mathcal{S}_{r,y}(s) = [1 - qM\{y(\beta)\}] \frac{p(\beta/r)}{p(s)}, \quad \frac{\partial \beta}{\partial s} = \frac{r}{1 + ry'(\beta)},$$

the chain rule first gives

$$\begin{aligned} \frac{\partial}{\partial s} \log \mathcal{S}_{r,y}(s) &= \frac{-qM'\{y(\beta)\}y'(\beta)}{1 - qM\{y(\beta)\}} \frac{\partial \beta}{\partial s} \\ &\quad + \frac{p'(\beta/r)}{p(\beta/r)} \frac{1}{r} \frac{\partial \beta}{\partial s} - \frac{p'(s)}{p(s)}. \end{aligned}$$

Using $-p'(x)/p(x) = \lambda(x)$, this becomes

$$\begin{aligned} \frac{\partial}{\partial s} \log \mathcal{S}_{r,y}(s) &= -\frac{qM'\{y(\beta)\}y'(\beta)}{1 - qM\{y(\beta)\}} \frac{r}{1 + ry'(\beta)} \\ &\quad - \frac{\lambda(\beta/r)}{1 + ry'(\beta)} + \lambda(s). \end{aligned} \tag{113}$$

The compact containment of β in $(0, a_q)$, together with (167) in Lemma A.1, implies

$$\inf_{\substack{y \in I \\ |rs - \alpha_q(y)| \leq \delta}} [1 - qM\{y(\beta)\}] > 0, \quad \sup_{\substack{y \in I \\ |rs - \alpha_q(y)| \leq \delta}} (|M'\{y(\beta)\}| + y'(\beta)) < \infty.$$

Thus the first term on the right side of (113) is $O(r)$ uniformly. For the remaining two terms, (171) in Lemma A.2 and $s \asymp \beta/r \asymp r^{-1}$ give

$$\lambda(s) = s + O(r), \quad \lambda(\beta/r) = \frac{\beta}{r} + O(r)$$

uniformly. The threshold equation $g_r(\beta) = r(s + y)$ is equivalent to

$$\frac{\beta}{r} = s + y - y(\beta).$$

Consequently,

$$\begin{aligned} \lambda(s) - \frac{\lambda(\beta/r)}{1 + ry'(\beta)} &= s - \frac{\beta/r}{1 + ry'(\beta)} + O(r) \\ &= \frac{rsy'(\beta) + y(\beta) - y}{1 + ry'(\beta)} + O(r) = O(1), \end{aligned}$$

because rs , y , $y(\beta)$, and $y'(\beta)$ are uniformly bounded. Hence

$$\sup_{y \in I} \sup_{|rs - \alpha_q(y)| \leq \delta} \left| \frac{\partial}{\partial s} \log \mathcal{S}_{r,y}(s) \right| = O(1). \tag{114}$$

Moreover, $\beta_{r,y,s}/r - s \geq \delta_I$ by (106), so $\beta_{r,y,s}/r \geq s + \delta_I$. Since p is decreasing on $[0, \infty)$, and since (167) in Lemma A.1 gives $0 \leq 1 - qM\{y(\beta_{r,y,s})\} \leq 1$, we have

$$0 \leq \mathcal{S}_{r,y}(s) \leq \frac{p(s + \delta_I)}{p(s)}.$$

Applying (178) in Lemma A.2 with $x = s$ and $\delta = \delta_I$ gives

$$0 \leq \mathcal{S}_{r,y}(s) \leq \frac{p(s + \delta_I)}{p(s)} \leq (1 + s^{-2}) \exp \left\{ -s\delta_I - \frac{\delta_I^2}{2} \right\} = o(r), \quad (115)$$

uniformly on the interval. The last $o(r)$ follows because (108) in Lemma 2.15 and $0 < \delta < m_I/2$ imply

$$s \geq \frac{\alpha_q(y) - \delta}{r} \geq \frac{m_I}{2r}$$

whenever $y \in I$ and $|rs - \alpha_q(y)| \leq \delta$, and therefore

$$r^{-1}(1 + s^{-2}) \exp \left\{ -s\delta_I - \frac{\delta_I^2}{2} \right\} \leq r^{-1}(1 + o(1)) \exp \left\{ -\frac{m_I\delta_I}{2r} - \frac{\delta_I^2}{2} \right\} \rightarrow 0$$

uniformly. Since $\partial_s \mathcal{S}_{r,y}(s) = \mathcal{S}_{r,y}(s) \partial_s \log \mathcal{S}_{r,y}(s)$, (114) and (115) imply

$$\sup_{y \in I} \sup_{|rs - \alpha_q(y)| \leq \delta} \left| \frac{\partial}{\partial s} \mathcal{S}_{r,y}(s) \right| = o(r). \quad (116)$$

For the null term, put

$$x_{\pm} = \frac{s \pm ry}{\sqrt{1 - r^2}}, \quad Q_{\pm} = \frac{\bar{\Phi}(x_{\pm})}{\bar{\Phi}(s)}.$$

Then

$$\frac{R_{0,r,-y}\{p(s)\}}{p(s)} = \frac{Q_+ + Q_-}{2}, \quad \frac{\partial Q_{\pm}}{\partial s} = Q_{\pm} \left\{ \lambda(s) - \frac{\lambda(x_{\pm})}{\sqrt{1 - r^2}} \right\}.$$

Equation (50) in Lemma 2.11, together with (172) in Lemma A.2, gives uniformly for $y \in I$ and $|rs - \alpha_q(y)| \leq \delta$

$$Q_{\pm} = \exp \left\{ -\frac{(rs)^2}{2} \mp rsy \right\} + o(1), \quad \frac{1}{r} \left\{ \lambda(s) - \frac{\lambda(x_{\pm})}{\sqrt{1 - r^2}} \right\} = -rs \mp y + o(1).$$

Therefore

$$\frac{1}{r} \frac{\partial}{\partial s} \mathcal{N}_{r,y}(s) = qe^{-(rs)^2/2} \{-rs \cosh(rsy) + y \sinh(rsy)\} + o(1) = q \frac{\partial}{\partial a} L_a(y) \Big|_{a=rs} + o(1). \quad (117)$$

By (108) and (110) in Lemma 2.15, the pairs (rs, y) under consideration range over a compact subset of $\{(a, y) : y \in I, 0 < a < a(y)\}$. Continuity, (101), and Lemma A.1 therefore show that $\partial_a L_a(y)|_{a=rs}$ is bounded away from zero uniformly. Combining (116) and (117) yields

$$\inf_{y \in I} \inf_{|rs - \alpha_q(y)| \leq \delta} \frac{\partial}{\partial s} \bar{R}_{r,-y}(s) > 0$$

for all sufficiently small r . Proposition 2.5 shows that $C_r(-y)$ is regular. The expression for $D(z)$ in Theorem 2.8, the limit $rC_r(-y) \rightarrow \alpha_q(y)$, and (49) in Lemma 2.11 give

$$D_r(-y) \rightarrow qL_{\alpha_q(y)}(y) = 1,$$

where the equality follows from (100). □

2.6.5 Conditional limit of D_r

The following lemma establishes the limit of $D_r(z)$ as $r \downarrow 0$, with D_* defined in Section 2.2.

Lemma 2.16. *For every $z \notin \{-1, -y_q\}$,*

$$D_r(z) \longrightarrow D_*(z). \quad (118)$$

Consequently,

$$\mathbb{E}[D_r(Z)] \longrightarrow \mathbb{E}[D_*(Z)] = \ell_=(q). \quad (119)$$

Proof. If $z > -1$, Lemma 2.12 gives $D_r(z) \rightarrow q$. If $-y_q < z < -1$, put $y = -z$ and apply Lemma 2.14. If $z < -y_q$, apply Lemma 2.15. This proves (118).

At every regular crossing, $0 \leq D_r \leq 1$; the value assigned at nonregular points also lies in $[0, 1]$. Dominated convergence gives the convergence in (119). Finally,

$$\begin{aligned} \mathbb{E}[D_*(Z)] &= q\mathbb{P}(Z > -1) + q \int_{-y_q}^{-1} M(-z)\phi(z) dz + \mathbb{P}(Z < -y_q) \\ &= q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q) \\ &= \ell_=(q). \end{aligned}$$

□

2.7 Proofs of Theorem 2.1 and Propositions 2.2 and 2.3

Proof of Theorem 2.1. Fix $\delta > 0$. Choose a compact set

$$E \subset \mathbb{R} \setminus \{-1, -y_q\}$$

such that

$$\int_E D_*(z)\phi(z) dz > \ell_=(q) - \frac{\delta}{4}.$$

The proofs of Lemmas 2.12, 2.14, and 2.15 give, for all sufficiently small r , brackets

$$A_r(z) < B_r(z), \quad z \in E,$$

and a constant $\gamma > 0$, such that

$$\sup_{z \in E} \sup_{u_q \leq c \leq A_r(z)} \bar{R}_{r,z}(c) \leq 1 - \gamma, \quad \inf_{z \in E} \bar{R}_{r,z}\{B_r(z)\} \geq 1 + \gamma,$$

$$\sup_{z \in E} \sup_{A_r(z) \leq c \leq B_r(z)} |\bar{R}_{0,r,z}(c) - D_*(z)| \leq \frac{\delta}{8}.$$

Fix such an r and put $C_* = \sup_{z \in E} B_r(z) < \infty$. The DKW argument used in Theorem 2.8, with these brackets in place of the regularity brackets, implies that for all sufficiently large N ,

$$\mathbb{E}_{\Omega_N}[\text{FDR}_{\Omega_N}(\text{BH}_q)] > \ell_=(q) - \frac{\delta}{2}.$$

For such an N , some realization ω^* satisfies

$$\text{FDR}_{\omega^*}(\text{BH}_q) > \ell_=(q) - \frac{\delta}{2}.$$

By Lemma 2.9, choose $\eta > 0$ so small that

$$\text{FDR}_{\omega^*}^{(\eta)}(\text{BH}_q) > \text{FDR}_{\omega^*}(\text{BH}_q) - \frac{\delta}{2} > \ell_{\leq}(q) - \delta. \quad (120)$$

The perturbed vector $T^{(\eta)}$ has finite dimension N , mean vector $(\theta_i)_{i=1}^N \in \mathbb{R}^N$, and covariance matrix $\Sigma_{\omega^*, \eta}$, which has unit diagonal and is strictly positive definite by (32) in Lemma 2.9. Thus the bound (120) establishes (5). Since $\delta > 0$ was arbitrary, taking the supremum proves (6). $\square$

Proof of Proposition 2.2. Recall from Lemma A.1 that, for $y > 1$, $a(y)$ is the unique positive maximizer of $a \mapsto L_a(y)$, equivalently the unique positive solution of

$$a(y) = y \tanh\{a(y)y\}.$$

By the monotonicity of $a(\cdot)$ established in that lemma, $a(y) \geq a(2) > 0$ for $y \geq 2$, so $a(y)y \rightarrow \infty$ and

$$\frac{a(y)}{y} = \tanh\{a(y)y\} \rightarrow 1.$$

Moreover,

$$0 \leq y - a(y) = y[1 - \tanh\{a(y)y\}] \leq 2ye^{-2a(y)y} \rightarrow 0.$$

Because $a(y)$ is the maximizer defining $M(y)$, we have $M(y) = L_{a(y)}(y)$. Substituting the definitions of $L_{a(y)}(y)$ and $\phi(y)$, expanding $\cosh(t) = (e^t + e^{-t})/2$, and completing the two squares gives

$$\begin{aligned} M(y)\phi(y) &= e^{-a(y)^2/2} \cosh\{a(y)y\} \frac{e^{-y^2/2}}{\sqrt{2\pi}} \\ &= \frac{1}{2\sqrt{2\pi}} \left[e^{-\{a(y)^2+y^2\}/2+a(y)y} + e^{-\{a(y)^2+y^2\}/2-a(y)y} \right] \\ &= \frac{1}{2\sqrt{2\pi}} \left[e^{-\{y-a(y)\}^2/2} + e^{-\{y+a(y)\}^2/2} \right]. \end{aligned} \quad (121)$$

Consequently, $M(y)\phi(y) \rightarrow 1/(2\sqrt{2\pi})$. The difference $M(y)\phi(y) - 1/(2\sqrt{2\pi})$ is integrable on $[1, \infty)$. Indeed, for all sufficiently large y , $a(y) \geq y/2$, so

$$y - a(y) \leq 2ye^{-y^2},$$

while

$$\left| e^{-\{y-a(y)\}^2/2} - 1 \right| \leq \frac{\{y-a(y)\}^2}{2}, \quad e^{-\{y+a(y)\}^2/2} \leq e^{-y^2/2}.$$

It follows from the definition of c_ℓ that, as $Y \rightarrow \infty$,

$$\int_1^Y M(y)\phi(y) dy = \frac{Y}{2\sqrt{2\pi}} + c_\ell - \Phi(1) + o(1). \quad (122)$$

Next, (121) gives

$$M(y) = \frac{1}{2}e^{y^2/2} \left[e^{-\{y-a(y)\}^2/2} + e^{-\{y+a(y)\}^2/2} \right] = \frac{1}{2}e^{y^2/2} \{1 + o(1)\}.$$

Since $M(y_q) = 1/q \rightarrow \infty$, (165) in Lemma A.1 implies $y_q \rightarrow \infty$. The identity $qM(y_q) = 1$ therefore gives

$$y_q^2 = 2 \log(1/q) + 2 \log 2 + o(1), \quad y_q = \sqrt{2 \log(1/q) + o(1)}. \quad (123)$$

Equations (183) in Lemma A.2 and (123) give

$$\bar{\Phi}(y_q) \leq \frac{\phi(y_q)}{y_q} = O\left(\frac{q}{y_q}\right) = o(q). \quad (124)$$

Substituting (122), (123), and (124) into (4) yields

$$\begin{aligned} \ell_=(q) &= q\Phi(1) + q \left\{ \frac{y_q}{2\sqrt{2\pi}} + c_\ell - \Phi(1) + o(1) \right\} + o(q) \\ &= \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q). \end{aligned}$$

Numerical quadrature of the absolutely convergent integral defining c_ℓ gives $c_\ell = 0.6492828\dots$ $\square$

Proof of Proposition 2.3. Since $q \in (0, 1)$, (3) gives $M(y_q) = 1/q > 1$. By Lemma A.1, $M = 1$ on $[0, 1]$ and M is strictly increasing on $(1, \infty)$, so $y_q > 1$ and $M(y) \geq 1$ for $1 \leq y \leq y_q$. Using (4),

$$\begin{aligned} \ell_=(q) - q &= q\Phi(1) + q \int_1^{y_q} M(y)\phi(y) dy + \bar{\Phi}(y_q) \\ &\quad - q \left\{ \Phi(1) + \int_1^{y_q} \phi(y) dy + \bar{\Phi}(y_q) \right\} \\ &= q \int_1^{y_q} \{M(y) - 1\}\phi(y) dy + (1 - q)\bar{\Phi}(y_q). \end{aligned}$$

The integral is nonnegative, and $(1 - q)\bar{\Phi}(y_q) > 0$ because $q < 1$ and $y_q < \infty$. Hence $\ell_=(q) > q$. $\square$

3 Lower bound for one-sided tests

3.1 Main result

For $0 < q < 1$, define

$$y_q^+ = \sqrt{2\log(1/q)} \quad (125)$$

and

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + \bar{\Phi}(y_q^+). \quad (126)$$

The following theorem gives the finite-dimensional one-sided Gaussian lower bound.

Theorem 3.1. *For every $q \in (0, 1)$ and $\delta > 0$, there are $m < \infty$, $\theta \in \mathbb{R}^m$, and a positive definite correlation matrix Σ such that BH at level q , applied to the one-sided p -values $P_i = \bar{\Phi}(T_i)$ from $T \sim N(\theta, \Sigma)$, satisfies*

$$\text{FDR}_{\theta, \Sigma}(\text{BH}_q) > \ell_{\leq}(q) - \delta.$$

Every true null in the construction has $\theta_i = 0$, and every nonnull has $\theta_i > 0$.

The first proposition gives the small- q expansion of the one-sided lower bound.

Proposition 3.2. *As $q \downarrow 0$,*

$$\ell_{\leq}(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

In particular, $\ell_{\leq}(q)/q \rightarrow \infty$.

The next proposition shows strict anticonservativeness at every nontrivial nominal level.

Proposition 3.3. *For every $q \in (0, 1)$, $\ell_{\leq}(q) > q$.*

Figure 2 in the Introduction plots $\ell_{\leq}(q) - q$ and $\ell_{\leq}(q)/q$ over the grid $q = 0.005k$, $k = 1, \dots, 200$. Table 3 gives the absolute inflation $\ell_{\leq}(q) - q$ and relative inflation $\ell_{\leq}(q)/q$ for commonly used nominal levels.

Table 3: The explicit one-sided lower bound and its inflation factor.

q	0.01	0.05	0.10	0.20
$\ell_{\leq}(q)$	0.01831	0.08101	0.15155	0.27955
$\ell_{\leq}(q)/q$	1.8311	1.6203	1.5155	1.3977

3.2 Roadmap

Fix q . For a small $r > 0$, let $(\Theta_i, B_i)_{i=1}^N$ be i.i.d. from a law $\nu_{q,r}^+$, independently of $Z, \varepsilon_1, \dots, \varepsilon_N$, which are independent standard normal variables, and set

$$T_i = \Theta_i + B_i Z + \sqrt{1 - B_i^2} \varepsilon_i. \quad (127)$$

The law $\nu_{q,r}^+$ has one null atom and two nonnull components. We first analyze the population BH crossing conditional on $Z = z$. Its conditional FDP converges, as $r \downarrow 0$, to

$$D_+^*(z) = \begin{cases} q, & z < 0, \\ qe^{z^2/2}, & 0 < z < y_q^+, \\ 1, & z > y_q^+. \end{cases} \quad (128)$$

The boundary points are immaterial because they have zero probabilities. Therefore

$$\mathbb{E}\{D_+^*(Z)\} = \frac{q}{2} + q \int_0^{y_q^+} e^{z^2/2} \phi(z) dz + \bar{\Phi}(y_q^+) = \ell_{\leq}(q).$$

We reuse the machinery and use similar notation in Subsections 2.3 and 2.4. Throughout this section, put $p_+(x) = \bar{\Phi}(x)$ and $u_s^+ = p_+^{-1}(s)$. Use the symbols

$$\pi_0^+, R_z^+, R_{0,z}^+, \bar{R}_z^+, \bar{R}_{0,z}^+, C^+(z), D^+(z)$$

as the one-sided analogues of the quantities in (7) and (10)–(15). The only change to (8)–(9) is

$$F_{\theta,b,z}^+(s) = \bar{\Phi}\left(\frac{u_s^+ - \theta - bz}{\sqrt{1 - b^2}}\right), \quad |b| < 1, \quad (129)$$

while $F_{\theta,b,z}^+(s) = \mathbf{1}\{\theta + bz \geq u_s^+\}$ for $|b| = 1$. We retain the empty-crossing convention $C^+(z) = \infty$, $D^+(z) = 0$, and the regularity definition in Definition 2.4.

The proofs of Lemmas 2.6 and 2.7 and Theorem 2.8 are p-value agnostic and apply after (129); the one-sided rejection set in Lemma 2.7 is $\{i : T_i \geq \hat{C}_N\}$. We also use the localized DKW bracketing in the proof of Theorem 2.8. Finally, the perturbation (30)–(31) and Lemma 2.9 apply to the one-sided p-values as well, because their continuity argument does not use the two-sided form of p .

3.3 Construction of $\nu_{q,r}$

This is the one-sided, sign-reversed specialization of the three-component construction in Subsection 2.5: the signal offset $y(a)$ is replaced by a , the envelope term $M\{y(a)\}$ by $e^{a^2/2}$, and the nonnull loading $+1$ by -1 .

Take $r > 0$, and set

$$b_r^+ = r p_+^{-1}(r^2). \quad (130)$$

We henceforth assume r is sufficiently small that $r^2 < 1/2$ and $b_r^+ < y_q^+$. For $b_r^+ \leq a \leq y_q^+$, define

$$W_r^+(a) = \frac{\{1 - qe^{a^2/2}\}p_+(a/r)}{q}. \quad (131)$$

Both factors in the numerator are nonnegative and decreasing in a , and the first vanishes at $a = y_q^+$. Hence W_r^+ is continuous and decreasing. Put

$$\tau_r^+ = W_r^+(b_r^+), \quad \mu_r^+([a, y_q^+]) = \frac{W_r^+(a)}{\tau_r^+}, \quad b_r^+ \leq a \leq y_q^+. \quad (132)$$

Thus μ_r^+ is a probability measure on $[b_r^+, y_q^+]$. Since $p_+(b_r^+/r) = r^2$, the quantile expansion (176) in Lemma A.2 gives $b_r^+ \rightarrow 0$, and hence

$$\tau_r^+ \sim \frac{1-q}{q} r^2. \quad (133)$$

Finally, set

$$\pi_{0,r}^+ = 1 - r - \tau_r^+, \quad p_+(c_r^+) = \frac{q(r + \tau_r^+)}{1 - q\pi_{0,r}^+}. \quad (134)$$

For all sufficiently small r , these quantities are well-defined and $\pi_{0,r}^+ > 0$.

The law $\nu_{q,r}^+$ is displayed in Table 4. Equivalently, for bounded measurable f ,

$$\int f \, d\nu_{q,r}^+ = \pi_{0,r}^+ f(0, r) + r f(c_r^+ + b_r^+, -1) + \tau_r^+ \int_{b_r^+}^{y_q^+} f(a/r + a, -1) \mu_r^+(da). \quad (135)$$

Table 4: One-sided population construction. The opposite loading signs are essential.

Type	Mass	Θ	B
Nulls	$\pi_{0,r}^+$	0	r
Primary nonnulls	r	$c_r^+ + b_r^+$	-1
Secondary nonnulls	$\tau_r^+ \mu_r^+(da)$	$a/r + a$	$-1, a \in [b_r^+, y_q^+]$

The nulls become more significant when Z is positive. Every nonnull statistic is deterministic conditional on Z and decreases with Z . The construction is not PRDS. To see this directly, pair a null i with a nonnull j , and fix $v \in (0, 1)$. Conditional on $P_i = s$, the event $\{P_j \geq v\}$, which is increasing in the p-values, has probability

$$\Phi\left(\frac{p_+^{-1}(v) - \theta_j + r p_+^{-1}(s)}{\sqrt{1 - r^2}}\right),$$

strictly decreasing in s . This also exhibits the negative null-to-nonnul covariance, opposite to the nonnegative covariance condition that guarantees Gaussian one-sided PRDS [Benjamini and Yekutieli, 2001, Chi et al., 2025]. On the other hand, all null–null correlations equal r^2 , so the null p-values are positively regression dependent and the full vector is PRDN in the terminology of Su [2018].

3.4 Population analysis

3.4.1 Preliminaries

The next lemma gives the one-sided counterparts of (47)–(48).

Lemma 3.4. *As $r \downarrow 0$,*

$$b_r^+ \rightarrow 0, \quad b_r^+/r \rightarrow \infty, \quad c_r^+ \rightarrow \infty, \quad r c_r^+ \rightarrow 0, \quad b_r^+/r - c_r^+ \rightarrow \infty. \quad (136)$$

Proof. By (130) and the quantile expansion (176) in Lemma A.2,

$$\frac{b_r^+}{r} = p_+^{-1}(r^2) \sim 2\sqrt{\log(1/r)} \rightarrow \infty, \quad b_r^+ \sim 2r\sqrt{\log(1/r)} \rightarrow 0.$$

Moreover, (133) gives $\tau_r^+ = O(r^2)$ and hence $\pi_{0,r}^+ \rightarrow 1$. Therefore (134) yields

$$p_+(c_r^+) \sim \frac{qr}{1-q}.$$

Another application of (176) in Lemma A.2 gives

$$c_r^+ \sim \sqrt{2 \log(1/r)}.$$

Thus $c_r^+ \rightarrow \infty$ and $rc_r^+ \rightarrow 0$, while

$$\frac{b_r^+}{r} - c_r^+ \sim (2 - \sqrt{2})\sqrt{\log(1/r)} \rightarrow \infty.$$

□

The next lemma gives the only new one-sided tail limit needed below.

Lemma 3.5. *For fixed $x > 0$ and bounded z, h , as $r \downarrow 0$,*

$$\frac{p_+\{(x/r + h - rz)/\sqrt{1-r^2}\}}{p_+(x/r + h)} \rightarrow \exp\left(xz - \frac{x^2}{2}\right). \quad (137)$$

The convergence is uniform on compact sets.

Proof. If $c = x/r + h$, then

$$\frac{c - rz}{\sqrt{1-r^2}} - c = r\left(\frac{x}{2} - z\right) + o(r),$$

and the local tail-ratio expansion in Lemma A.2 proves (137). □

Apply the reused population notation to $\nu_{q,r}^+$, and write the resulting quantities as $R_{r,z}^+$, $R_{0,r,z}^+$, $\bar{R}_{r,z}^+$, $\bar{R}_{0,r,z}^+$, $C_r^+(z)$, and $D_r^+(z)$. Then

$$\begin{aligned} \bar{R}_{r,z}^+(c) &= \bar{R}_{0,r,z}^+(c) + \frac{qr}{p_+(c)} \mathbf{1}\{c_r^+ + b_r^+ - z \geq c\} \\ &\quad + \frac{q\tau_r^+}{p_+(c)} \int_{b_r^+}^{y_q^+} \mathbf{1}\{a/r + a - z \geq c\} \mu_r^+(da), \end{aligned} \quad (138)$$

$$\bar{R}_{0,r,z}^+(c) = q\pi_{0,r}^+ \frac{p_+\{(c - rz)/\sqrt{1-r^2}\}}{p_+(c)}. \quad (139)$$

3.4.2 The regime $z < 0$

The next lemma analyzes the crossing generated by the primary block when $z < 0$.

Lemma 3.6. *For every compact $K \subset (-\infty, 0)$, uniformly for $z \in K$,*

$$C_r^+(z) = c_r^+ + o\{(c_r^+)^{-1}\}. \quad (140)$$

and

$$D_r^+(z) \rightarrow q. \quad (141)$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (140). Fix a compact $K \subset (-\infty, 0)$. Uniformly for $z \in K$ and bounded h , all primary and secondary nonnulls are rejected at $c = c_r^+ + h/c_r^+$. The primary statistic equals $c_r^+ + b_r^+ - z$, so its distance above the cutoff is $b_r^+ - z - h/c_r^+$. This converges to $-z > 0$ because $b_r^+ \rightarrow 0$, $c_r^+ \rightarrow \infty$, and h is bounded. Among the secondary nonnulls, the smallest statistic is attained at $a = b_r^+$. Its distance above the cutoff is

$$\frac{b_r^+}{r} + b_r^+ - z - c_r^+ - \frac{h}{c_r^+} = \left(\frac{b_r^+}{r} - c_r^+ \right) + b_r^+ - z - \frac{h}{c_r^+},$$

which tends to infinity by (136) in Lemma 3.4. Thus both gaps are positive for all sufficiently small r , uniformly when z ranges over a compact subset of $(-\infty, 0)$ and h remains bounded.

Set $c = c_r^+ + h/c_r^+$. Since every nonnull is rejected, the population curve decomposes as

$$\bar{R}_{r,z}^+(c) = \bar{R}_{0,r,z}^+(c) + \frac{q(r + \tau_r^+)}{p_+(c)}.$$

For the null contribution, (136) in Lemma 3.4 gives $c \rightarrow \infty$ and $rc \rightarrow 0$. Define the change in the null cutoff by

$$\Delta_r = \frac{c - rz}{\sqrt{1 - r^2}} - c.$$

The expansion $(1 - r^2)^{-1/2} = 1 + r^2/2 + O(r^4)$ gives

$$\begin{aligned} c\Delta_r &= c^2 \left\{ \frac{1}{\sqrt{1 - r^2}} - 1 \right\} - \frac{rzc}{\sqrt{1 - r^2}} \\ &= \frac{1}{2}(rc)^2 - z(rc) + o\{(rc)^2\} + o(rc) \rightarrow 0. \end{aligned}$$

Recall that $\lambda(x) = \phi(x)/\bar{\Phi}(x)$. Since $z < 0$, we also have $\Delta_r > 0$, and hence

$$0 \leq -\log \frac{p_+(c + \Delta_r)}{p_+(c)} = \int_c^{c + \Delta_r} \lambda(t) dt \leq \Delta_r \left\{ c + \Delta_r + (c + \Delta_r)^{-1} \right\}.$$

The final bound follows from (171) in Lemma A.2 and tends to zero because $c\Delta_r \rightarrow 0$ and $\Delta_r \rightarrow 0$. Exponentiating yields

$$\frac{p_+\{(c - rz)/\sqrt{1 - r^2}\}}{p_+(c)} = \frac{p_+(c + \Delta_r)}{p_+(c)} \rightarrow 1.$$

Because $\pi_{0,r}^+ \rightarrow 1$, (139) now yields $\bar{R}_{0,r,z}^+(c) \rightarrow q$. For the combined nonnull contribution, the calibration (134) gives the exact identity

$$\frac{q(r + \tau_r^+)}{p_+(c_r^+ + h/c_r^+)} = (1 - q\pi_{0,r}^+) \frac{p_+(c_r^+)}{p_+(c_r^+ + h/c_r^+)}.$$

The last ratio tends to e^h by (180) in Lemma A.2, whereas $1 - q\pi_{0,r}^+ \rightarrow 1 - q$. Combining the two contributions gives

$$\bar{R}_{r,z}^+(c_r^+ + h/c_r^+) \rightarrow q + (1 - q)e^h, \quad \bar{R}_{0,r,z}^+(c_r^+ + h/c_r^+) \rightarrow q. \quad (142)$$

The first limit is below one for $h < 0$, equals one at zero, and is above one for $h > 0$. We next exclude earlier crossings. By (136) in Lemma 3.4, uniformly for $z \in K$, all nonnulls are rejected when $u_q^+ \leq c \leq c_r^+$, and $(c - rz)/\sqrt{1 - r^2} > c$ for small r . Hence

$$\bar{R}_{r,z}^+(c) < q\pi_{0,r}^+ + \frac{q(r + \tau_r^+)}{p_+(c)} \leq q\pi_{0,r}^+ + \frac{q(r + \tau_r^+)}{p_+(c_r^+)} = 1.$$

For every fixed $\epsilon > 0$, the first limit in (142) is larger than one at $h = \epsilon$ for all sufficiently small r , uniformly on K . Therefore

$$c_r^+ < C_r^+(z) \leq c_r^+ + \frac{\epsilon}{c_r^+}, \quad z \in K.$$

Since $\epsilon > 0$ is arbitrary,

$$C_r^+(z) = c_r^+ + o\{(c_r^+)^{-1}\}$$

uniformly on K .

Proof of (141). By Proposition 2.5, it remains to prove that the derivative of $\bar{R}_{r,z}^+(c_r^+ + h/c_r^+)$ with respect to h is positive there. Write $\lambda(x) = \phi(x)/\bar{\Phi}(x)$. At $c = c_r^+ + h/c_r^+$, the derivatives with respect to h of the null and combined nonnull contributions are

$$\bar{R}_{0,r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) \frac{\lambda(c) - \frac{1}{\sqrt{1-r^2}}\lambda\{(c-rz)/\sqrt{1-r^2}\}}{c_r^+} \quad \text{and} \quad \frac{q(r + \tau_r^+)}{p_+(c)} \frac{\lambda(c)}{c_r^+}.$$

To evaluate the first derivative, fix $H < \infty$. Uniformly for $z \in K$ and $|h| \leq H$, (136) in Lemma 3.4 gives

$$\frac{c}{c_r^+} \longrightarrow 1, \quad \frac{c - rz}{c_r^+ \sqrt{1-r^2}} \longrightarrow 1.$$

The hazard bounds (171) in Lemma A.2 then imply

$$\frac{\lambda(c)}{c_r^+} \longrightarrow 1, \quad \frac{1}{c_r^+ \sqrt{1-r^2}} \lambda\left(\frac{c - rz}{\sqrt{1-r^2}}\right) \longrightarrow 1$$

uniformly on the same set. Hence, uniformly for $z \in K$ and $|h| \leq H$,

$$\frac{1}{c_r^+} \left[\lambda(c) - \frac{1}{\sqrt{1-r^2}} \lambda\left(\frac{c - rz}{\sqrt{1-r^2}}\right) \right] \longrightarrow 0, \quad \bar{R}_{0,r,z}^+(c) \longrightarrow q,$$

where the second limit is from (142). Therefore

$$\frac{\partial}{\partial h} \bar{R}_{0,r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) \longrightarrow 0$$

uniformly on the same set.

For the combined nonnull contribution, the calibration (134) rewrites its derivative exactly as

$$(1 - q\pi_{0,r}^+) \frac{p_+(c_r^+)}{p_+(c_r^+ + h/c_r^+)} \frac{\lambda(c_r^+ + h/c_r^+)}{c_r^+}.$$

Here $1 - q\pi_{0,r}^+ \rightarrow 1 - q$, the middle factor converges uniformly to e^h by (180) in Lemma A.2, and the final factor converges uniformly to one by (171) in Lemma A.2. Together with the null-derivative limit above, uniformly for $z \in K$ and $|h| \leq H$,

$$\frac{\partial}{\partial h} \frac{q(r + \tau_r^+)}{p_+(c_r^+ + h/c_r^+)} \longrightarrow (1 - q)e^h, \quad \frac{\partial}{\partial h} \bar{R}_{r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) \longrightarrow (1 - q)e^h.$$

Since $\inf_{|h| \leq H} (1 - q)e^h = (1 - q)e^{-H} > 0$, for all sufficiently small r ,

$$\inf_{z \in K} \inf_{|h| \leq H} \frac{\partial}{\partial h} \bar{R}_{r,z}^+\left(c_r^+ + \frac{h}{c_r^+}\right) > 0.$$

Proposition 2.5 therefore proves regularity uniformly on K .

Now put

$$h_r(z) = c_r^+ \{C_r^+(z) - c_r^+\}.$$

The cutoff limit (140) gives $h_r(z) \rightarrow 0$ uniformly on K , and

$$C_r^+(z) = c_r^+ + \frac{h_r(z)}{c_r^+}.$$

In particular, $|h_r(z)| \leq 1$ uniformly on K for all sufficiently small r . The second limit in (142) is uniform for $z \in K$ and h in compact sets. Evaluating it at $h = h_r(z)$ therefore gives

$$\sup_{z \in K} \left| \bar{R}_{0,r,z}^+ \left(c_r^+ + \frac{h_r(z)}{c_r^+} \right) - q \right| \rightarrow 0.$$

Using the expression for $D(z)$ in Theorem 2.8, we conclude that, uniformly on K ,

$$D_r^+(z) = \bar{R}_{0,r,z}^+ \{C_r^+(z)\} \rightarrow q.$$

□

3.4.3 The regime $z \in (0, y_q^+)$

The remaining two regimes use three common bounds. Let $S_{r,z}^+(c)$ denote the secondary contribution, the last term in (138), and put

$$d = \frac{c + z}{1 + r}.$$

The rejection boundary for the secondary index is $a = rd$. If $rd \leq b_r^+$, then $q\tau_r^+ \leq p_+(b_r^+/r) \leq p_+(d)$; if $b_r^+ < rd < y_q^+$, then

$$q\tau_r^+ \mu_r^+([rd, y_q^+]) = \{1 - qe^{r^2 d^2/2}\} p_+(d) \leq p_+(d);$$

and if $rd \geq y_q^+$, the contribution is zero. Thus

$$0 \leq S_{r,z}^+(c) \leq \frac{p_+(d)}{p_+(c)}. \quad (143)$$

On a compact subset of $z > 0$, (134), (136) in Lemma 3.4, and (181) in Lemma A.2 show that the primary contribution, whenever nonzero, is bounded by

$$\frac{qr}{p_+(c_r^+ + b_r^+ - z)} = o(1) \quad (144)$$

uniformly. Finally, the same calculation used for (137) in Lemma 3.5 gives, whenever $c \rightarrow \infty$ and $rc \rightarrow x \geq 0$,

$$\bar{R}_{0,r,z}^+(c) \rightarrow q \exp \left(xz - \frac{x^2}{2} \right). \quad (145)$$

The next lemma analyzes the crossing generated by the secondary block when $0 < z < y_q^+$.

Lemma 3.7. *For every compact $K \subset (0, y_q^+)$, uniformly for $z \in K$,*

$$C_r^+(z) = \frac{z}{r} + o(1). \quad (146)$$

and

$$D_r^+(z) \rightarrow qe^{z^2/2}. \quad (147)$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (146). Fix a compact $K \subset (0, y_q^+)$. Uniformly for $z \in K$ and bounded h , (136) in Lemma 3.4 implies that at $c = z/r + h$, the primary nonnull is not rejected. A secondary nonnull is rejected if

$$a \geq \beta_r := z + \frac{rh}{1+r}.$$

Consequently, (131)–(132) and the local tail-ratio expansion in Lemma A.2 yield

$$\frac{q\tau_r^+ \mu_r^+([\beta_r, y_q^+])}{p_+(z/r + h)} \longrightarrow \{1 - qe^{z^2/2}\}e^{zh}.$$

Combining this with (137) in Lemma 3.5 gives

$$\bar{R}_{r,z}^+(z/r + h) \longrightarrow qe^{z^2/2} + \{1 - qe^{z^2/2}\}e^{zh}, \quad \bar{R}_{0,r,z}^+(z/r + h) \longrightarrow qe^{z^2/2}. \quad (148)$$

Because $qe^{z^2/2} < 1$ uniformly away from one on K , the first limit crosses one strictly at $h = 0$.

We now exclude earlier crossings. Fix $\epsilon > 0$. If a crossing existed at some $c \leq z/r - \epsilon$ along sequences $r \downarrow 0$ and $z \in K$, pass to subsequences so that $z \rightarrow z_0$ and $rc \rightarrow x \in [0, z_0]$. For bounded c , the full curve tends to q . If $c \rightarrow \infty$ and $x < z_0$, then $d - c \rightarrow z_0 - x > 0$; hence (143)–(145) make the full curve tend to

$$q \exp\left(xz_0 - \frac{x^2}{2}\right) \leq qe^{z_0^2/2} < 1.$$

It remains only that $x = z_0$. Write $h = c - z/r \leq -\epsilon$. If h is bounded, (148) gives a limit strictly below one. If $h \rightarrow -\infty$, then

$$d - c = -\frac{rh}{1+r} > 0, \quad c(d - c) \sim -z_0 h \longrightarrow \infty.$$

Thus $p_+(d)/p_+(c) \rightarrow 0$, while the null curve tends to $qe^{z_0^2/2} < 1$, a contradiction. Hence no crossing occurs at or before $z/r - \epsilon$. The upper bracket $z/r + \epsilon$ follows from (148). Letting $\epsilon \downarrow 0$ gives

$$C_r^+(z) = \frac{z}{r} + o(1)$$

uniformly on K .

Proof of (147). It remains to prove regularity. Put $c = z/r + h$ and $\beta = z + rh/(1+r)$, and write $\lambda = \phi/\bar{\Phi}$. The derivative of the null contribution with respect to h is

$$\bar{R}_{0,r,z}^+(c) \left[\lambda(c) - \frac{1}{\sqrt{1-r^2}} \lambda\{(c-rz)/\sqrt{1-r^2}\} \right],$$

and the logarithmic derivative of the secondary contribution is

$$-\frac{qr\beta e^{\beta^2/2}}{(1+r)\{1 - qe^{\beta^2/2}\}} + \lambda(c) - \frac{\lambda(\beta/r)}{1+r}.$$

The hazard bounds (171) in Lemma A.2, (136) in Lemma 3.4, and the tail ratios already used show uniformly on compact h -windows that the derivative of the full curve converges to

$$z\{1 - qe^{z^2/2}\}e^{zh} > 0.$$

The primary term is absent in this window, and continuity of W_r^+ excludes a jump in the secondary term. Together with the first-crossing exclusion, this verifies Definition 2.4 uniformly on K .

Finally, put $h_r = C_r^+(z) - z/r$. Then $h_r \rightarrow 0$ uniformly on K , so the compact-uniform second limit in (148) gives

$$D_r^+(z) = \bar{R}_{0,r,z}^+\{C_r^+(z)\} \longrightarrow qe^{z^2/2}.$$

□

3.4.4 The regime $z > y_q^+$

For $z > y_q^+$, define

$$\alpha_q^+(z) = z - \sqrt{z^2 - (y_q^+)^2} \in (0, z).$$

The next lemma analyzes the crossing generated by the null curve in this regime.

Lemma 3.8. *For every compact $K \subset (y_q^+, \infty)$, uniformly for $z \in K$,*

$$rC_r^+(z) \longrightarrow \alpha_q^+(z). \quad (149)$$

and

$$D_r^+(z) \longrightarrow 1. \quad (150)$$

Moreover, for all sufficiently small r , $C_r^+(z)$ is regular for every $z \in K$.

Proof. Proof of (149). Fix a compact $K \subset (y_q^+, \infty)$. For $z \in K$ and $0 < x < z$, (136) in Lemma 3.4 implies that the primary nonnull is not rejected at $c = x/r$. The secondary boundary equals

$$\frac{x + rz}{1 + r} = x + r(z - x) + O(r^2) > x,$$

so its normalized contribution vanishes exponentially. Equation (137) in Lemma 3.5 gives

$$\bar{R}_{r,z}^+(x/r) \longrightarrow q \exp\left(xz - \frac{x^2}{2}\right), \quad (151)$$

with the same limit for the null curve. The equation

$$q \exp\left(xz - \frac{x^2}{2}\right) = 1$$

has smaller positive solution $\alpha_q^+(z)$. The limit in (151) is below one for $x < \alpha_q^+(z)$ and above one for $\alpha_q^+(z) < x < z$.

We now exclude earlier crossings. Fix $\epsilon > 0$ uniformly small enough that

$$0 < \alpha_q^+(z) - \epsilon < \alpha_q^+(z) + \epsilon < z, \quad z \in K.$$

If a crossing existed at some $c \leq \{\alpha_q^+(z) - \epsilon\}/r$, pass to subsequences so that $z \rightarrow z_0$ and $rc \rightarrow x$ with $0 \leq x \leq \alpha_q^+(z_0) - \epsilon$. Equations (143)–(145) then give either q , when $x = 0$, or

$$q \exp\left(xz_0 - \frac{x^2}{2}\right) < 1,$$

uniformly away from one. At $c = \{\alpha_q^+(z) + \epsilon\}/r$, the null contribution alone is larger than one for all sufficiently small r , uniformly on K . Therefore

$$\alpha_q^+(z) - \epsilon < rC_r^+(z) < \alpha_q^+(z) + \epsilon.$$

Letting $\epsilon \downarrow 0$ proves

$$rC_r^+(z) \longrightarrow \alpha_q^+(z)$$

uniformly on K .

Proof of (150). It remains to prove regularity. Write $\lambda = \phi/\bar{\Phi}$. Uniformly near $c = \alpha_q^+(z)/r$,

$$\frac{d}{dc} \log \frac{p_+\{(c - rz)/\sqrt{1 - r^2}\}}{p_+(c)} = \lambda(c) - \frac{1}{\sqrt{1 - r^2}} \lambda\{(c - rz)/\sqrt{1 - r^2}\} = r(z - rc) + o(r) > 0.$$

The secondary term and its derivative are exponentially smaller, and the primary term is absent. Continuity of W_r^+ excludes a secondary jump. Thus the full curve is strictly increasing through the localized crossing. Together with the first-crossing exclusion, this verifies Definition 2.4 uniformly on K .

At the population cutoff, (136) in Lemma 3.4 implies that the primary term is zero, and, with $d = \{C_r^+(z) + z\}/(1+r)$, (143) tends to zero because $d - C_r^+(z) \rightarrow z - \alpha_q^+(z) > 0$. The null term consequently tends to one, and regularity gives

$$D_r^+(z) = \bar{R}_{0,r,z}^+ \{C_r^+(z)\} \longrightarrow 1.$$

□

3.5 Proofs of the one-sided lower-bound results

Proof of Theorem 3.1. Choose a compact set

$$E \subset \mathbb{R} \setminus \{0, y_q^+\}$$

such that

$$\mathbb{E}\{D_+^*(Z)\mathbf{1}\{Z \in E\}\} > \ell_{\leq}(q) - \frac{\delta}{4}.$$

The proofs of Lemmas 3.6, 3.7, and 3.8 give, for all sufficiently small r , brackets

$$A_r(z) < B_r(z), \quad z \in E,$$

and a constant $\gamma > 0$ such that

$$\sup_{z \in E} \sup_{u_q^+ \leq c \leq A_r(z)} \bar{R}_{r,z}^+(c) \leq 1 - \gamma, \quad \inf_{z \in E} \bar{R}_{r,z}^+\{B_r(z)\} \geq 1 + \gamma,$$

$$\sup_{z \in E} \sup_{A_r(z) \leq c \leq B_r(z)} \left| \bar{R}_{0,r,z}^+(c) - D_+^*(z) \right| \leq \frac{\delta}{8}.$$

Fix such an r , and put $C_* = \sup_{z \in E} B_r(z) < \infty$. For $z \in E$, define

$$\mathcal{E}_N(z) = \max \left\{ \|\hat{R}_N - R_{r,z}^+\|_{\infty}, \|\hat{R}_{0,N} - \pi_{0,r}^+ R_{0,r,z}^+\|_{\infty} \right\}$$

and

$$t_* = \frac{p_+(C_*)}{q} \min \left\{ \gamma, \frac{\delta}{8} \right\}.$$

Following the same argument in the proof of Theorem 2.8, Lemma 2.6 gives

$$\sup_{z \in E} \mathbb{P}\{\mathcal{E}_N(z) > t_* \mid Z = z\} \leq 4e^{-2Nt_*^2}.$$

On the complementary event, the empirical and population curves differ by at most $q\mathcal{E}_N(z)/p_+(C_*)$ throughout $[u_q^+, C_*]$. The two population brackets and Lemma 2.7 therefore imply

$$A_r(z) < \hat{C}_N \leq B_r(z), \quad \text{FDP}_N \geq D_+^*(z) - \frac{\delta}{4}.$$

Consequently,

$$\begin{aligned} \mathbb{E}_{\Omega_N} \{\text{FDR}_{\Omega_N}(\text{BH}_q)\} &= \mathbb{E}_{\Omega_N, Z, \varepsilon}(\text{FDP}_N) \\ &\geq \int_E \left\{ D_+^*(z) - \frac{\delta}{4} \right\} \phi(z) \, dz - 4e^{-2Nt_*^2} \\ &> \ell_{\leq}(q) - \frac{\delta}{2} - 4e^{-2Nt_*^2}. \end{aligned}$$

For all sufficiently large N , this gives

$$\mathbb{E}_{\Omega_N}\{\text{FDR}_{\Omega_N}(\text{BH}_q)\} > \ell_{\leq}(q) - \frac{3\delta}{4}.$$

Therefore some deterministic realization ω satisfies the same inequality. The one-sided analogue of Lemma 2.9 permits an $\eta > 0$ for which the perturbed model has a positive definite correlation matrix and

$$\text{FDR}_{\omega}^{(\eta)}(\text{BH}_q) > \ell_{\leq}(q) - \delta.$$

All means and null labels are unchanged by the perturbation. Equation (31) shows that its off-diagonal covariances are $(1 - \eta^2)b_i b_j$, so the asserted pre-perturbation sign pattern follows from Table 4. $\square$

Proof of Proposition 3.2. Since $e^{-(y_q^+)^2/2} = q$, the Mills expansion in Lemma A.2 gives

$$\bar{\Phi}(y_q^+) \sim \frac{q}{\sqrt{2\pi} y_q^+} = o(q).$$

Substitute $y_q^+ = \sqrt{2\log(1/q)}$ into (126). $\square$

Proof of Proposition 3.3. Put

$$M_+(z) = \sup_{a \geq 0} \exp\left(az - \frac{a^2}{2}\right) = \begin{cases} 1, & z \leq 0, \\ e^{z^2/2}, & z > 0. \end{cases}$$

Then (128) can be written as $D_+^*(z) = \min\{qM_+(z), 1\}$. It equals q for $z \leq 0$ and is strictly larger than q for every $z > 0$. Hence $\ell_{\leq}(q) = \mathbb{E}\{D_+^*(Z)\} > q$. $\square$

4 Upper bounds

4.1 Upper bound for two-sided tests

In this section, we examine a class of common-factor models that contains the construction in Subsection 2.5. We show that the upper bound is $O(q\sqrt{\log(1/q)})$, which gives matching orders for small q .

Theorem 4.1. *Let*

$$T_i = \mu_i + a_i Z + \sqrt{1 - a_i^2} \varepsilon_i, \quad 1 \leq i \leq m,$$

where $Z, \varepsilon_1, \dots, \varepsilon_m$ are i.i.d. standard normal random variables and $|a_i| \leq 1$. Form the two-sided Gaussian p -values $P_i = p(|T_i|)$ and apply BH at level $0 < q \leq 2\bar{\Phi}(1) \approx 0.3173$. Let $\mathcal{H}_0 = \{i : \mu_i = 0\}$, $m_0 = |\mathcal{H}_0|$, and $\pi_0 = m_0/m$. If u_q is defined by $2\bar{\Phi}(u_q) = q$, then $\text{FDR}(\text{BH}_q) = 0$ when $m_0 = 0$, and otherwise

$$\text{FDR}(\text{BH}_q) \leq q \left\{ 1 + \pi_0 \left(\frac{1-q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) \right\}.$$

Consequently, the worst-case FDR in this common-factor class is $O\{q\sqrt{\log(1/q)}\}$, uniformly in m , the means $(\mu_1, \dots, \mu_m)$, and the loadings $(a_1, \dots, a_m)$, as $q \downarrow 0$.

Remark 4.2 (PRDN for the two-sided null p -values). The theorem imposes no sign restriction on the loadings a_i , so the null correlations $a_i a_j$ may be negative. Nevertheless, the two-sided null p -values satisfy PRDN. Let $T_0 = (T_i : i \in \mathcal{H}_0)$ be the null subvector, let $a_0 = (a_i : i \in \mathcal{H}_0)$, and put

$$D_0 = \text{diag}(1 - a_i^2 : i \in \mathcal{H}_0).$$

The null covariance matrix is

$$\Sigma_0 = D_0 + a_0 a_0^\top.$$

Assume first that $|a_i| < 1$ for every null coordinate. Let

$$B = \text{diag}\{\text{sign}(a_i) : i \in \mathcal{H}_0\}, \quad r = |a_0|,$$

with $\text{sign}(0) = 1$. Then

$$B\Sigma_0 B = D_0 + rr^\top.$$

By the Sherman–Morrison formula,

$$(D_0 + rr^\top)^{-1} = D_0^{-1} - \frac{D_0^{-1}rr^\top D_0^{-1}}{1 + r^\top D_0^{-1}r}.$$

Thus every off-diagonal entry of $(D_0 + rr^\top)^{-1}$ is nonpositive. The Karlin–Rinott absolute-value multinormal MTP₂ criterion [Karlin and Rinott, 1981] therefore applies to BT_0 , and $(|(BT_0)_i| : i \in \mathcal{H}_0)$ is MTP₂. Since $|(BT_0)_i| = |T_i|$, coordinatewise sign changes do not alter the two-sided p -values $p(|T_i|)$.

If some $|a_i| = 1$, define $a_i^{(\eta)} = (1 - \eta)a_i$ for all $i \in \mathcal{H}_0$, and let $\eta \downarrow 0$. The corresponding absolute null vectors converge weakly to $(|T_i| : i \in \mathcal{H}_0)$. Under the measure-theoretic definition of MTP₂, the class of MTP₂ probability measures is closed under weak convergence [Colangelo et al., 2006]. Hence the limiting absolute-value law remains MTP₂. Appendix A.1 of Su [2018] then implies that the induced two-sided null p -values satisfy PRDN.

Proof. A leave-one-out form of BH self-consistency is standard in FDR proofs [Benjamini and Yekutieli, 2001]; we include the argument for completeness. If $m_0 = 0$, then $V = 0$ identically and hence FDR = 0. Assume henceforth that $m_0 \geq 1$. Fix a true null i . Let R_i be the BH rejection count after replacing its p -value by zero; in particular, $R_i \geq 1$. Pointwise in the p -value vector,

$$\frac{\mathbf{1}\{i \text{ is rejected}\}}{R \vee 1} = \frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i}.$$

Indeed, if i is rejected, replacing P_i by zero does not change the step-up rejection count. Conversely, if $P_i \leq qR_i/m$, restoring P_i leaves at least R_i values at or below qR_i/m . Thus at least R_i values remain below the BH threshold qR/m , so BH makes at least R_i rejections. Coordinatewise monotonicity gives at most R_i rejections. Here ties cause no ambiguity: the BH rejection set is $\{j : P_j \leq qR/m\}$ and has exactly R elements. If more than R values were at most qR/m , then $P_{(R+1)} \leq qR/m < q(R+1)/m$, contradicting maximality of R . Summing gives the leave-one-out identity

$$\text{FDR} = \sum_{i \in \mathcal{H}_0} \mathbb{E} \left[\frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i} \right]. \quad (152)$$

Conditional on $Z = z$, P_i is independent of R_i . Indeed, set $s_j = \sqrt{1 - a_j^2}$ and let $\mathcal{R}(\cdot)$ denote the BH step-up rejection count. Then

$$R_i = \mathcal{R}(P_1, \dots, P_{i-1}, 0, P_{i+1}, \dots, P_m), \quad P_j = p(|\mu_j + a_j z + s_j \varepsilon_j|), \quad j \neq i.$$

Hence, conditional on $Z = z$, R_i is measurable with respect to $\sigma(\varepsilon_j : j \neq i, s_j > 0)$; coordinates with $s_j = 0$ are constants after conditioning. Meanwhile $P_i = p(|a_i z + s_i \varepsilon_i|)$ for a true null i , or is deterministic if $s_i = 0$. Since ε_i is independent of $\{\varepsilon_j : j \neq i\}$, the claim follows. With

$$F_{i,z}(t) = \mathbb{P}(P_i \leq t \mid Z = z),$$

and $0 < qR_i/m \leq q$,

$$\begin{aligned}
\mathbb{E} \left[\frac{\mathbf{1}\{P_i \leq qR_i/m\}}{R_i} \middle| Z = z \right] &= \mathbb{E} \left[\frac{1}{R_i} \mathbb{P} \left(P_i \leq \frac{qR_i}{m} \middle| Z = z, R_i \right) \middle| Z = z \right] \\
&= \mathbb{E} \left[\frac{F_{i,z}(qR_i/m)}{R_i} \middle| Z = z \right] \\
&= \frac{q}{m} \mathbb{E} \left[\frac{F_{i,z}(qR_i/m)}{qR_i/m} \middle| Z = z \right] \\
&\leq \frac{q}{m} \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t}.
\end{aligned} \tag{153}$$

Summing (153) over the true nulls gives

$$\mathbb{E}(\text{FDP} \mid Z = z) \leq \frac{q}{m} \sum_{i \in \mathcal{H}_0} \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t}. \tag{154}$$

To average (154) over Z , it remains to control

$$\int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz$$

for each true null i . We do this on $|z| \leq u_q$; the event $|Z| > u_q$ is handled at the end by $\text{FDP} \leq 1$. Fix a true null index i . Put $r = |a_i|$ and $s = \sqrt{1 - r^2}$, and first suppose $s > 0$. Conditional on $Z = z$, $T_i \sim N(a_i z, s^2)$. The inequality $t = p(u) = 2\bar{\Phi}(u) \leq q$ holds if and only if $u \geq u_q$, and in this case

$$F_{i,z}(p(u)) = \int_u^\infty s^{-1} \phi\{(x - a_i z)/s\} dx + \int_u^\infty s^{-1} \phi\{(x + a_i z)/s\} dx.$$

Since $s^2 = 1 - a_i^2$,

$$\begin{aligned}
\frac{s^{-1} \phi\{(x - a_i z)/s\}}{\phi(x)} &= \frac{1}{s} \exp \left\{ -\frac{(x - a_i z)^2}{2s^2} + \frac{x^2}{2} \right\} \\
&= \frac{1}{s} \exp \left\{ \frac{z^2}{2} - \frac{(a_i x - z)^2}{2s^2} \right\}.
\end{aligned}$$

The second tail is the same calculation with z replaced by $-z$. Choose $\eta_+, \eta_- \in \{-1, 1\}$ with $\{\eta_+, \eta_-\} = \{-1, 1\}$ so that $\eta_+ a_i z \geq 0$ and $\eta_- a_i z \leq 0$. For $x \geq u$,

$$|a_i x - \eta_+ z| \geq (ru - |z|)_+, \quad |a_i x - \eta_- z| \geq ru + |z|.$$

Both right-hand sides are nondecreasing in u , so for $u \geq u_q$,

$$\begin{aligned}
\phi(z) \frac{F_{i,z}(p(u))}{p(u)} &\leq \frac{\phi(z)}{p(u)} \frac{e^{z^2/2}}{s} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right] \int_u^\infty \phi(x) dx \\
&= \frac{1}{2s\sqrt{2\pi}} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right],
\end{aligned}$$

because $\int_u^\infty \phi(x) dx = p(u)/2$. Taking the supremum gives

$$\phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} \leq \frac{1}{2s\sqrt{2\pi}} \left[e^{-(ru_q - |z|)_+^2/(2s^2)} + e^{-(ru_q + |z|)^2/(2s^2)} \right]. \tag{155}$$

Integrating (155) over $|z| \leq u_q$ and using symmetry gives

$$\int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz \leq \frac{1}{s\sqrt{2\pi}} \left\{ \int_0^{u_q} e^{-(ru_q - z)_+^2/(2s^2)} dz + \int_0^{u_q} e^{-(ru_q + z)^2/(2s^2)} dz \right\}. \tag{156}$$

The two integrals on the right-hand side of (156) are

$$\begin{aligned}\int_0^{u_q} e^{-(ru_q-z)^2/(2s^2)} dz &= \int_0^{ru_q} e^{-y^2/(2s^2)} dy + (1-r)u_q, \\ \int_0^{u_q} e^{-(ru_q+z)^2/(2s^2)} dz &= \int_{ru_q}^{(1+r)u_q} e^{-y^2/(2s^2)} dy.\end{aligned}$$

Consequently, writing $\lambda = \sqrt{(1-r)/(1+r)} \in (0, 1]$,

$$\begin{aligned}\int_{-u_q}^{u_q} \phi(z) \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t} dz &\leq \frac{1}{s\sqrt{2\pi}} \left\{ \int_0^{(1+r)u_q} e^{-y^2/(2s^2)} dy + (1-r)u_q \right\} \\ &= \left\{ \Phi\left(\frac{(1+r)u_q}{s}\right) - \frac{1}{2} \right\} + \frac{(1-r)u_q}{s\sqrt{2\pi}} \\ &= \Phi(u_q/\lambda) - \frac{1}{2} + \frac{u_q\lambda}{\sqrt{2\pi}},\end{aligned}\tag{157}$$

where the final equality in (157) uses $s^2 = 1 - r^2 = (1-r)(1+r)$, so $(1+r)/s = 1/\lambda$ and $(1-r)/s = \lambda$. The right-hand side of (157) is increasing in $\lambda \in (0, 1]$. Its derivative is

$$\frac{u_q}{\sqrt{2\pi}} \left[1 - \lambda^{-2} \exp\{-u_q^2/(2\lambda^2)\} \right] \geq 0,$$

because $u_q \geq 1$ and, on putting $w = \lambda^{-2} \geq 1$,

$$we^{-u_q^2 w/2} \leq we^{-w/2} \leq 2/e < 1.$$

Hence (157) is at most

$$\Phi(u_q) - \frac{1}{2} + \frac{u_q}{\sqrt{2\pi}} = \frac{1-q}{2} + \frac{u_q}{\sqrt{2\pi}}.\tag{158}$$

When $r = 1$, the conditional p -value is deterministic. For $|z| < u_q$, it exceeds q , and the boundary $|z| = u_q$ has Lebesgue measure zero; hence (158) remains valid. When $r = 0$, the conditional null p -value is uniform and independent of Z ; (158) remains valid, although it is not sharp.

Let $A = \{|Z| \leq u_q\}$. Decomposing over A and A^c ,

$$\text{FDR} = \mathbb{E}[\text{FDP} \mathbf{1}_A] + \mathbb{E}[\text{FDP} \mathbf{1}_{A^c}].$$

For the first term, integrate (154) over $z \in [-u_q, u_q]$ and use (158). For the second term, no conditional tail bound is needed: since $0 \leq \text{FDP} \leq 1$,

$$\mathbb{E}[\text{FDP} \mathbf{1}_{A^c}] \leq \mathbb{P}(A^c) = \mathbb{P}(|Z| > u_q) = 2\bar{\Phi}(u_q) = q.\tag{159}$$

Combining (154), (158), and (159) yields

$$\text{FDR} \leq q + \frac{q}{m} \sum_{i \in \mathcal{H}_0} \left(\frac{1-q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) = q \left\{ 1 + \pi_0 \left(\frac{1-q}{2} + \frac{u_q}{\sqrt{2\pi}} \right) \right\},$$

as claimed. $\square$

4.2 Upper bounds for one-sided tests

The one-sided lower bound belongs to the rank-one common-factor class studied in Theorem 4.1. We now show that its leading constant cannot be improved within this class.

Theorem 4.3. *Let*

$$T_i = \theta_i + a_i Z + \sqrt{1 - a_i^2} \varepsilon_i, \quad 1 \leq i \leq m, \quad (160)$$

where $Z, \varepsilon_1, \dots, \varepsilon_m$ are independent standard normal random variables and $|a_i| \leq 1$. Test $H_i : \theta_i \leq 0$ with $P_i = \bar{\Phi}(T_i)$. Let $\mathcal{H}_0 = \{i : \theta_i \leq 0\}$, $m_0 = |\mathcal{H}_0|$, and $\pi_0 = m_0/m$. For $0 < q < 1/2$, put $u_q^+ = \bar{\Phi}^{-1}(q) > 0$. The FDR is zero when $m_0 = 0$, and otherwise

$$\text{FDR}(\text{BH}_q) \leq q \left\{ 2 + \pi_0 \left(\frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}} \right) \right\}. \quad (161)$$

Consequently, the worst-case FDR in this rank-one class is $O\{q\sqrt{\log(1/q)}\}$, uniformly in m , the means $(\theta_1, \dots, \theta_m)$, and the loadings $(a_1, \dots, a_m)$, as $q \downarrow 0$.

The following corollary identifies the exact leading constant for the rank-one class.

Corollary 4.4. *Let the supremum range over all finite rank-one models satisfying Theorem 4.3, with arbitrary means and scalar loadings. Then, as $q \downarrow 0$,*

$$\sup \text{FDR}(\text{BH}_q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + O(q).$$

Thus the constant $1/\sqrt{\pi}$ in Proposition 3.2 is optimal in the rank-one class.

Proof of Theorem 4.3. If there are no true nulls, the FDR is zero. Partition the true nulls into the two sign groups

$$\mathcal{H}_{0,1} = \{i \in \mathcal{H}_0 : a_i \geq 0\}, \quad \mathcal{H}_{0,-1} = \{i \in \mathcal{H}_0 : a_i < 0\}.$$

Let $\mathcal{S} \subseteq \{-1, 1\}$ index the nonempty groups. The leave-one-out identity (152) and conditional reduction (153) are p-value free. Conditional on the factor $Z = z$, they bound the contribution of a true null i by

$$\frac{q}{m} \sup_{0 < t \leq q} \frac{F_{i,z}(t)}{t}, \quad F_{i,z}(t) = \mathbb{P}(P_i \leq t \mid Z = z),$$

and the boundary mean $\theta_i = 0$ maximizes this conditional tail.

Suppose $i \in \mathcal{H}_{0,\sigma}$, where $\sigma \in \mathcal{S}$, and write $r = |a_i|$, $s = \sqrt{1 - r^2}$, and $Y_\sigma = \sigma Z \sim N(0, 1)$. The boundary-null conditional distribution given $Y_\sigma = y$ has CDF $F_{i,y}$. For $s > 0$, the one-tail Gaussian density-ratio calculation gives

$$\phi(y) \sup_{0 < t \leq q} \frac{F_{i,y}(t)}{t} \leq \frac{1}{s\sqrt{2\pi}} \exp \left\{ -\frac{(ru_q^+ - y)_+^2}{2s^2} \right\}. \quad (162)$$

Indeed, write $t = p_+(u)$, where $u \geq u_q^+$. The bivariate normal density identity gives

$$\phi(y) \frac{F_{i,y}\{p_+(u)\}}{p_+(u)} = \frac{1}{p_+(u)} \int_u^\infty \phi(x) \frac{1}{s} \phi\left(\frac{y - rx}{s}\right) dx \leq \frac{1}{s\sqrt{2\pi}} \exp \left\{ -\frac{(ru - y)_+^2}{2s^2} \right\}.$$

The right side is maximized over $u \geq u_q^+$ at $u = u_q^+$, proving (162). Integrating over $y \leq u_q^+$ yields

$$\begin{aligned} I(r) &:= \int_{-\infty}^{u_q^+} \phi(y) \sup_{0 < t \leq q} \frac{F_{i,y}(t)}{t} dy \\ &\leq \frac{1}{2} + \frac{(1-r)u_q^+}{s\sqrt{2\pi}} \\ &\leq \frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}}. \end{aligned} \quad (163)$$

The last inequality uses $(1-r)/s = \sqrt{(1-r)/(1+r)} \leq 1$. If $r = 1$, then on $Y_\sigma < u_q^+$ the boundary-null p-value is larger than q , so (163) remains valid.

Let V_σ be the false rejections in sign group σ and $\text{FDP}_\sigma = V_\sigma/(R \vee 1)$. On $A_\sigma = \{Y_\sigma \leq u_q^+\}$, sum the per-null conditional bound over $\mathcal{H}_{0,\sigma}$ and apply (163). This gives

$$\mathbb{E}(\text{FDP}_\sigma \mathbf{1}_{A_\sigma}) \leq q \frac{|\mathcal{H}_{0,\sigma}|}{m} \left(\frac{1}{2} + \frac{u_q^+}{\sqrt{2\pi}} \right).$$

On A_σ^c , use $0 \leq \text{FDP}_\sigma \leq 1$ and $\mathbb{P}(A_\sigma^c) = q$. Summing over $\sigma \in \mathcal{S}$, using $|\mathcal{S}| \leq 2$ and $\text{FDP} = \sum_{\sigma \in \mathcal{S}} \text{FDP}_\sigma$, proves (161). $\square$

Proof of Corollary 4.4. Theorem 3.1 gives the lower bound because the perturbation in (30) can still be written in the form (160), after combining its independent residual Gaussian terms. Theorem 4.3 gives the upper bound. Finally,

$$u_q^+ = \sqrt{2 \log(1/q)} + o(1),$$

so the upper and lower bounds have the same leading term and differ by at most $O(q)$. $\square$

5 Discussion

Theorem 2.1 and Proposition 2.2 show that two-sided Gaussian testing under unrestricted correlation can have worse guarantees than the nominal BH level suggests. In particular, the BH procedure can have FDR larger than its nominal level by a factor of order $\sqrt{\log(1/q)}$ as $q \downarrow 0$, so no universal constant multiple of q can bound its FDR over all Gaussian correlation matrices. Because the lower-bound construction belongs to the common-factor class, Theorem 4.1 shows that the worst-case FDR within this class is of exact order $q\sqrt{\log(1/q)}$ as $q \downarrow 0$. The leading constants do not match: Theorem 4.1 gives the upper constant $1/\sqrt{\pi}$, whereas Proposition 2.2 gives the lower constant $1/(2\sqrt{\pi})$. Thus the upper and lower constants differ by a factor of two. The one-sided results sharpen this picture. Theorem 3.1 and Proposition 3.2 give the larger leading constant $1/\sqrt{\pi}$, and Corollary 4.4 shows that this constant is optimal in the rank-one common-factor class. The upper bound allows arbitrary means and scalar loadings.

There are, however, other procedures with valid FDR guarantees for dependent two-sided Gaussian tests. One approach is the conditional-calibration framework of Fithian and Lei [2022]. The method decomposes FDR into hypothesis-wise contributions and, for each null i , conditions on a statistic that removes nuisance dependence while leaving a calibrated one-dimensional null distribution for the i th p -value. Their concrete dependence-adjusted BH procedure (dBH) calibrates a hypothesis-specific cutoff for the BH q -value and rejects H_i when its q -value falls below that calibrated cutoff. In general, dBH is at least as powerful as the Benjamini–Yekutieli procedure; under PRDS, it is at least as powerful as BH [Fithian and Lei, 2022]. Thus it provides finite-sample FDR control under known dependence without paying the full harmonic-factor correction of Benjamini and Yekutieli [2001].

Another route is the fixed- X knockoff filter of Barber and Candès [2015]. In a Gaussian linear model, knockoffs augment the design with synthetic variables that mimic the correlation structure of the original variables, and the resulting original-versus-knockoff comparisons create null statistics with the symmetry needed for FDR control. The correlated-Gaussian-testing interpretation is made explicit by Li and Fithian [2021], who recast fixed- X knockoffs as adding user-generated Gaussian noise to a Gaussian estimator to obtain a whitened estimator, followed by conditional inference. Knockoffs can use structural information and flexible feature-importance statistics unavailable to marginal testing. Their weakness is that whitening can require enough added noise to erase much of the signal when the original test statistics are strongly correlated, causing substantial power loss [Li and Fithian, 2021]. Knockoffs also face a threshold phenomenon when the rejection set is small. The calibrated knockoff method of Luo et al. [2025] uses conditional calibration to uniformly improve knockoff procedures, with especially large gains in small-rejection regimes.

More broadly, understanding FDR guarantees under dependence is essential because many real-world test statistics are well approximated by a multivariate Gaussian law. In such settings, the correlation matrix is often structured, but the induced two-sided p -values need not satisfy PRDS or the sign structure required by standard positive-dependence arguments. Two-sided testing can destroy monotonicity even when the underlying Gaussian variables have positive correlations. Procedures that explicitly use dependence information, and theory that clarifies when BH itself is reliable, are therefore both needed.

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A Auxiliary lemmas

This appendix collects the properties of the envelope and Gaussian tails used to define and analyze $\nu_{q,r}$. The following lemma characterizes the likelihood-ratio envelope.

Lemma A.1. *For $0 \leq y \leq 1$, $M(y) = 1$. For $y > 1$, $a \mapsto L_a(y)$ has a unique maximizer $a(y) > 0$, characterized by*

$$a(y) = y \tanh\{a(y)y\}. \quad (164)$$

The maps $a : (1, \infty) \rightarrow (0, \infty)$ and $M : (1, \infty) \rightarrow (1, \infty)$ are continuous and strictly increasing, with

$$\lim_{y \downarrow 1} a(y) = 0, \quad \lim_{y \rightarrow \infty} M(y) = \infty. \quad (165)$$

The inverse $y = y(a)$ is continuously differentiable on $(0, \infty)$, and

$$y'(a) = \frac{1 - y(a)^2 \operatorname{sech}^2\{ay(a)\}}{\tanh\{ay(a)\} + ay(a) \operatorname{sech}^2\{ay(a)\}} > 0. \quad (166)$$

Moreover, with $a_q = a(y_q)$, the deficit in the secondary survival mass satisfies

$$1 - qM\{y(a)\} > 0 \quad (0 < a < a_q), \quad 1 - qM\{y(a_q)\} = 0, \quad \frac{d}{da}[1 - qM\{y(a)\}] < 0. \quad (167)$$

Proof. Let

$$g_y(a) = \log L_a(y) = -\frac{a^2}{2} + \log \cosh(ay).$$

Then

$$g'_y(a) = -a + y \tanh(ay). \quad (168)$$

If $y \leq 1$, then $y \tanh(ay) \leq ay^2 \leq a$, so $M(y) = L_0(y) = 1$.

For $y > 1$, put $t = ay$. Equation $g'_y(a) = 0$ is equivalent to

$$y^2 = \frac{t}{\tanh t}. \quad (169)$$

To verify the strict inequality needed below, let $H(t) = \tanh t - t \operatorname{sech}^2 t$. Then $H(0) = 0$ and $H'(t) = 2t \operatorname{sech}^2 t \tanh t > 0$ for $t > 0$. Hence

$$\frac{d}{dt} \left(\frac{t}{\tanh t} \right) = \frac{\tanh t - t \operatorname{sech}^2 t}{\tanh^2 t} > 0.$$

Moreover, $t/\tanh t \rightarrow 1$ as $t \downarrow 0$ and $t/\tanh t \rightarrow \infty$ as $t \rightarrow \infty$. Because $y^2 > 1$, (169) therefore has a unique positive solution. The sign of g'_y changes from positive to negative there, so the solution is the unique maximizer. Moreover,

$$a(y) = \sqrt{t \tanh t}, \quad y = \sqrt{\frac{t}{\tanh t}}, \quad (170)$$

Both parameter functions in (170) are smooth and strictly increasing in $t > 0$. They tend respectively to zero and one as $t \downarrow 0$, and both tend to infinity as $t \rightarrow \infty$. Thus $a(y)$ is continuous and strictly increasing on $(1, \infty)$, with $a(y) \rightarrow 0$ as $y \downarrow 1$; the inverse function theorem applied to the second parameter function also shows that $a(y)$ is continuously differentiable. Since $M(y) = L_{a(y)}(y)$, the chain rule and $\partial_a L_{a(y)}(y) = 0$ give

$$M'(y) = M(y)a(y) \tanh\{a(y)y\} > 0.$$

Thus M is continuous and strictly increasing. Also

$$M(y) \geq L_y(y) \geq \frac{1}{2}e^{y^2/2} \rightarrow \infty.$$

Finally, apply the implicit function theorem to

$$F(a, y) = a - y \tanh(ay) = 0.$$

At the maximizer, with $t = ay > 0$, equation (169) gives

$$y^2 \operatorname{sech}^2(ay) = \frac{t \operatorname{sech}^2 t}{\tanh t} = \frac{2t}{\sinh(2t)} < 1.$$

Consequently,

$$F_a = 1 - y^2 \operatorname{sech}^2(ay) > 0, \quad F_y = -\tanh(ay) - ay \operatorname{sech}^2(ay) < 0.$$

Thus (166) in Lemma A.1 follows from $y' = -F_a/F_y$.

It remains to prove (167) in Lemma A.1. Since $a_q = a(y_q)$, we have $y(a_q) = y_q$, and therefore $1 - qM\{y(a_q)\} = 0$ by (3). If $0 < a < a_q$, then $y(a) < y_q$; since M is strictly increasing on $(1, \infty)$, $qM\{y(a)\} < qM(y_q) = 1$. Finally,

$$\frac{d}{da} [1 - qM\{y(a)\}] = -qM'\{y(a)\}y'(a) < 0,$$

because $M' > 0$ on $(1, \infty)$ and $y' > 0$ on $(0, \infty)$. □

The final lemma collects the Mills bounds and quantile asymptotics used to control Gaussian tails.

Lemma A.2. *The normal hazard $\lambda(x) = \phi(x)/\bar{\Phi}(x)$ is increasing on $\mathbb{R}$ and satisfies*

$$x < \lambda(x) < x + x^{-1} \quad (x > 0). \quad (171)$$

Moreover, as $x \rightarrow \infty$,

$$\lambda(x) = x + x^{-1} + O(x^{-3}). \quad (172)$$

As $x \rightarrow \infty$,

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} \{1 + o(1)\}. \quad (173)$$

For every fixed $0 < \sigma < 1$, as $x \rightarrow \infty$,

$$\frac{\bar{\Phi}(\sigma x)}{\bar{\Phi}(x)} \sim \frac{1}{\sigma} \exp \left\{ \frac{(1 - \sigma^2)x^2}{2} \right\}. \quad (174)$$

For every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$, as $x \rightarrow \infty$,

$$\frac{p(x+u)}{p(x+v)} = \frac{x+v}{x+u} \exp \left\{ -x(u-v) - \frac{u^2 - v^2}{2} \right\} \{1 + o(1)\}. \quad (175)$$

If $c > 0$ is fixed, then, as $q \downarrow 0$,

$$p^{-1}(cq) \sim \sqrt{2 \log(1/q)}. \quad (176)$$

In particular, for every fixed $c_1, c_2 > 0$,

$$\frac{p^{-1}(c_1 q)}{p^{-1}(c_2 q)} \longrightarrow 1 \quad (q \downarrow 0).$$

Finally, for $x \geq 0$,

$$2\bar{\Phi}(x) \leq e^{-x^2/2}. \quad (177)$$

For $x > 0$ and $\delta \geq 0$,

$$\frac{p(x+\delta)}{p(x)} \leq \left(1 + x^{-2}\right) \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\}. \quad (178)$$

Consequently, if $x \rightarrow \infty$, then the following tail-ratio limits hold.

(i) For every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$,

$$\log \frac{p(x+u)}{p(x+v)} = -x(u-v) - \frac{u^2 - v^2}{2} + o(1). \quad (179)$$

(ii) For every $0 < H < \infty$, uniformly over $|h| \leq H$,

$$\frac{p(x)}{p(x+h/x)} \longrightarrow e^h. \quad (180)$$

(iii) If $0 < \delta_0 \leq \delta \leq \delta_1 < \infty$, then

$$\frac{p(x+\delta)}{p(x)} \longrightarrow 0 \quad (181)$$

uniformly in δ .

(iv) If $\delta_x \geq 0$ and $x\delta_x \rightarrow \infty$, then

$$\frac{p(x+\delta_x)}{p(x)} \longrightarrow 0. \quad (182)$$

Proof. For $x > 0$, the upper Mills bound follows from

$$\bar{\Phi}(x) < \frac{1}{x} \int_x^\infty t\phi(t) \, dt = \frac{\phi(x)}{x}. \quad (183)$$

Integration by parts gives

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} - \int_x^\infty \frac{\phi(t)}{t^2} dt > \frac{\phi(x)}{x} - \frac{\bar{\Phi}(x)}{x^2},$$

which proves the lower Mills bound

$$\frac{x}{1+x^2}\phi(x) < \bar{\Phi}(x). \quad (184)$$

A further integration by parts gives

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} - \frac{\phi(x)}{x^3} + 3 \int_x^\infty \frac{\phi(t)}{t^4} dt,$$

where

$$0 < \int_x^\infty \frac{\phi(t)}{t^4} dt < \frac{\phi(x)}{x^5}.$$

Hence

$$\bar{\Phi}(x) = \frac{\phi(x)}{x} \{1 - x^{-2} + O(x^{-4})\},$$

and inversion proves (172) in Lemma A.2. Equations (183) and (184) imply (171) in Lemma A.2 for $x > 0$. Since $\lambda(x) - x > 0$ also holds trivially for $x \leq 0$, the identity

$$\lambda'(x) = \lambda(x)\{\lambda(x) - x\}$$

shows that λ is increasing on $\mathbb{R}$.

Equations (183) and (184) give (173) in Lemma A.2. Applying this at x and σx proves (174) in Lemma A.2. Applying (173) in Lemma A.2 at $x + u$ and $x + v$ gives (175) in Lemma A.2, for every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$. Moreover, as $x \rightarrow \infty$,

$$-\log p(x) = \frac{x^2}{2} + O(\log x),$$

which yields (176) in Lemma A.2 by inversion. To prove $2\bar{\Phi}(x) \leq e^{-x^2/2}$, set

$$d(x) = e^{-x^2/2} - 2\bar{\Phi}(x).$$

Then $d(0) = 0$, $\lim_{x \rightarrow \infty} d(x) = 0$, and

$$d'(x) = e^{-x^2/2} \left(\sqrt{\frac{2}{\pi}} - x \right).$$

Thus d first increases and then decreases to zero, so $d(x) \geq 0$ for all $x \geq 0$. Finally, (183) and (184) imply

$$\frac{p(x+\delta)}{p(x)} \leq \frac{1+x^2}{x(x+\delta)} \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\} \leq (1+x^{-2}) \exp \left\{ -x\delta - \frac{\delta^2}{2} \right\},$$

which proves (178) in Lemma A.2.

It remains to prove the four stated tail-ratio consequences. Taking logarithms in (175) in Lemma A.2 gives (179) in Lemma A.2, because

$$\log \frac{x+v}{x+u} = o(1)$$

for every $0 < H < \infty$, uniformly over $|u| \vee |v| \leq H$. For (180) in Lemma A.2, apply (175) in Lemma A.2 with $u = 0$ and $v = h/x$. For every $0 < H < \infty$, uniformly over $|h| \leq H$,

$$\frac{p(x)}{p(x+h/x)} = \frac{x+h/x}{x} \exp \left\{ h + \frac{h^2}{2x^2} \right\} \{1 + o(1)\} \longrightarrow e^h.$$

For (181) in Lemma A.2, apply (179) in Lemma A.2 with $u = \delta$ and $v = 0$. If $0 < \delta_0 \leq \delta \leq \delta_1 < \infty$, then

$$\log \frac{p(x + \delta)}{p(x)} \leq -x\delta_0 + O(1) \longrightarrow -\infty$$

uniformly in δ . Finally, (182) in Lemma A.2 follows from (178) in Lemma A.2 with $\delta = \delta_x$, because $x\delta_x \rightarrow \infty$. $\square$